Introduction to Machine Learning

Mahesh Mohan M R
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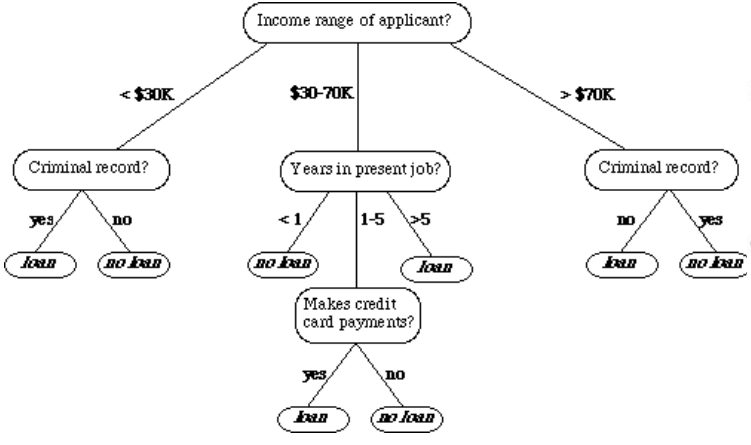
Robotic Manufacturing



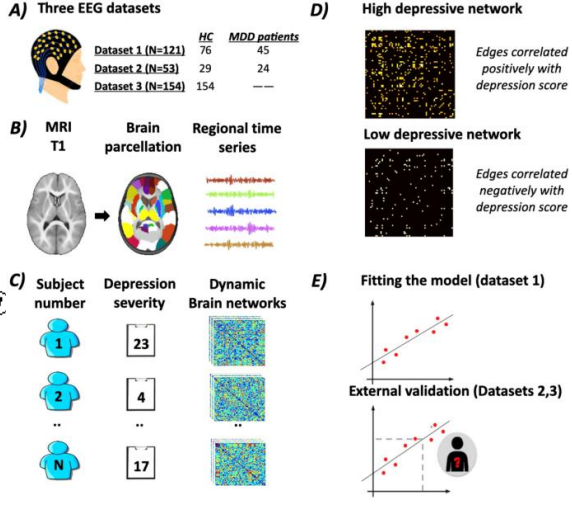
AI for Agriculture



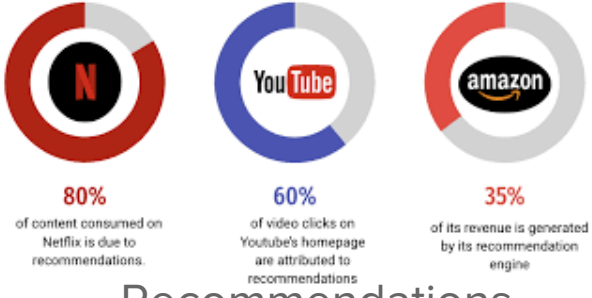
Smart Diagnosis



Loan Approval



Depression Classification

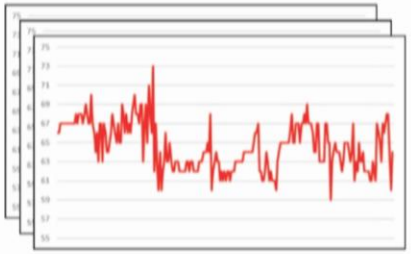


Recommendations

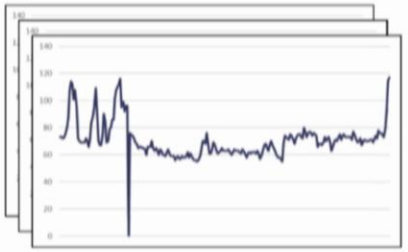


Time Series Data (1 dimensional)





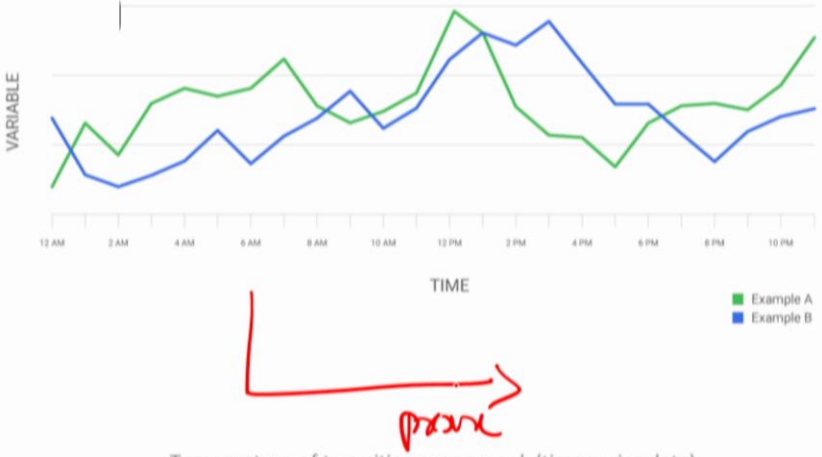
ECG

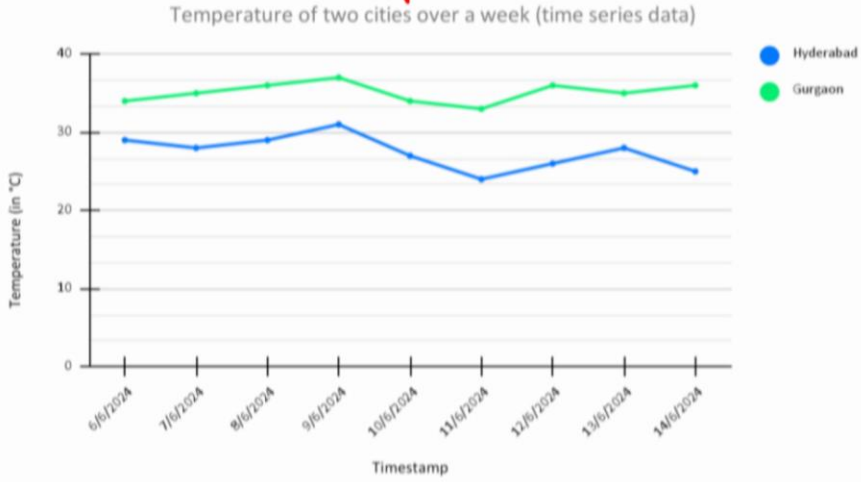


Heart rate - IoT

IoT device





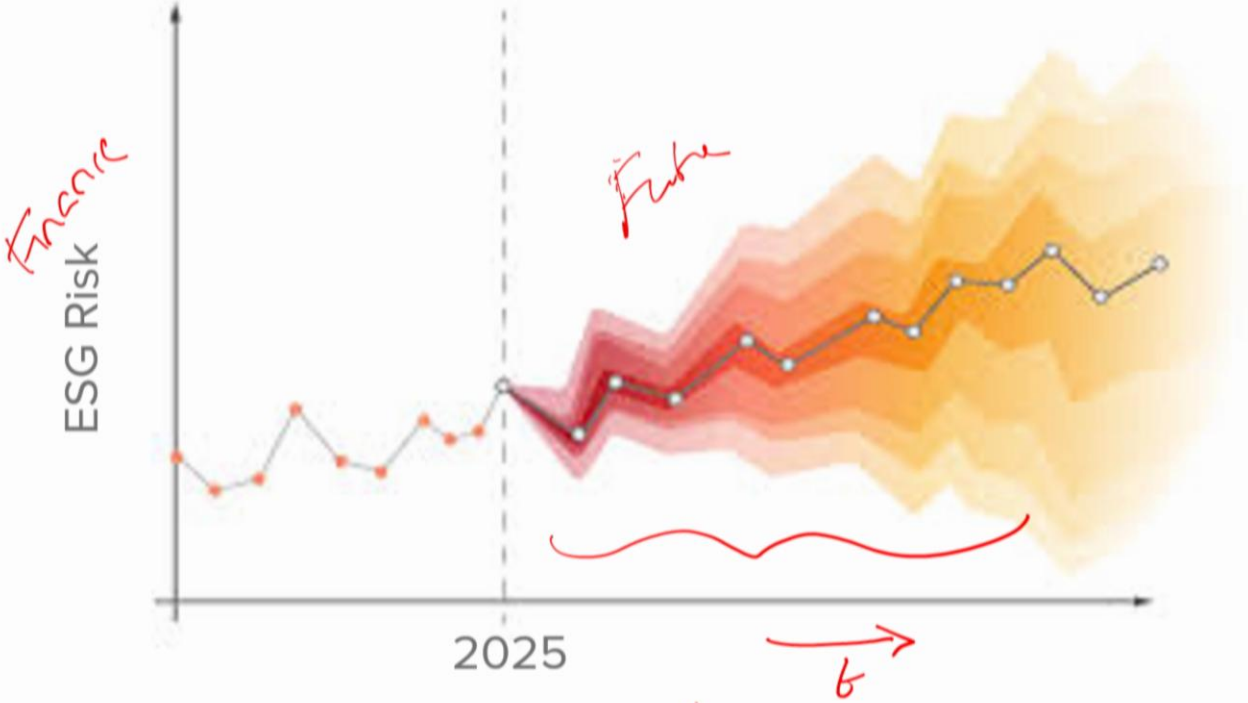


Time Series Data (1 dimensional)

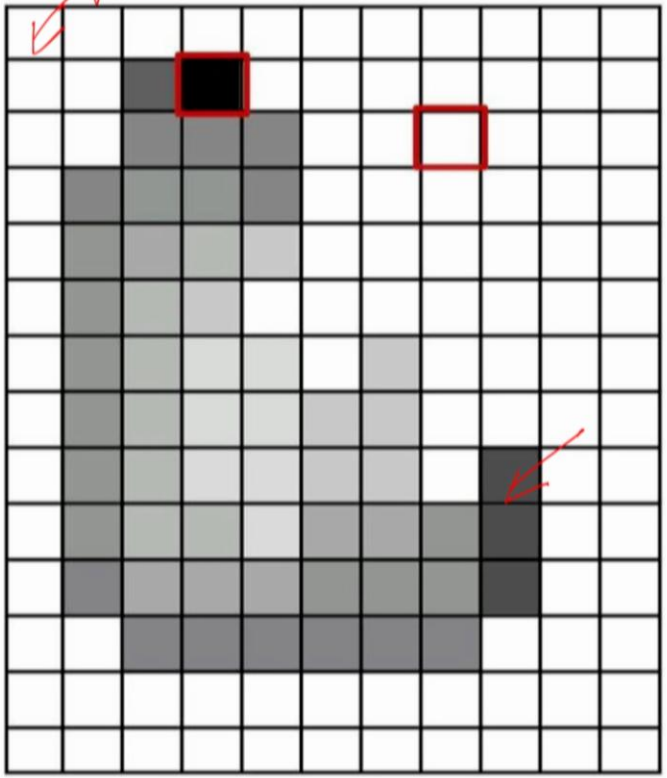
ICML
International Conference
On Machine Learning



Wonky profession



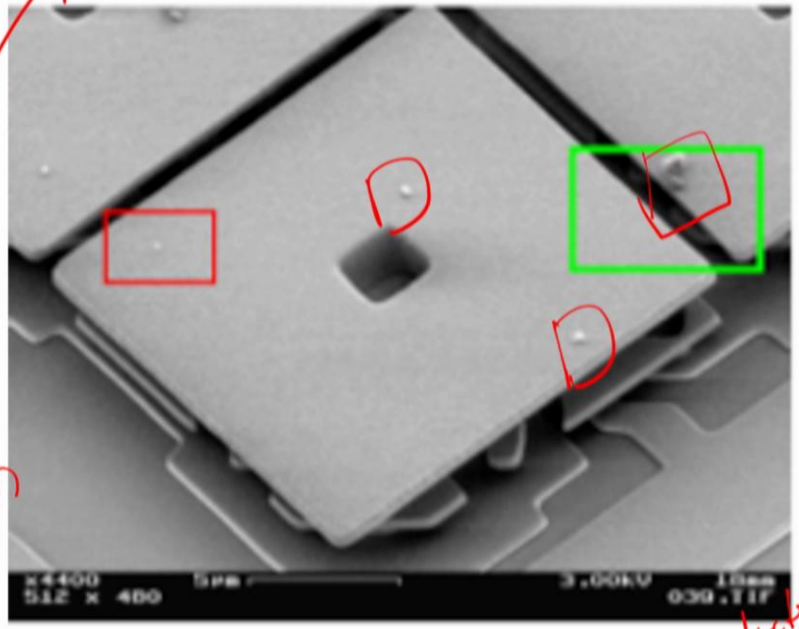
Time Series Data (1 dimensional)



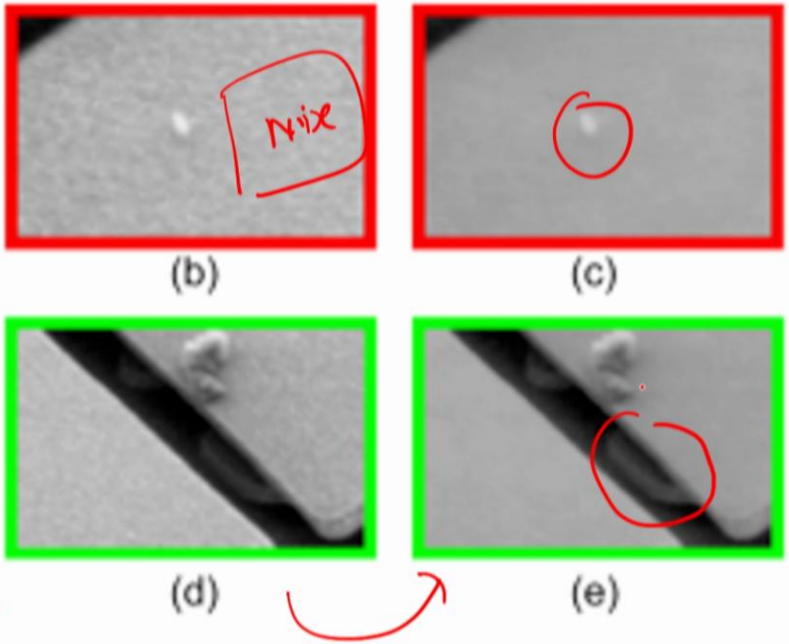
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255	127	145	200	200	175	175	95	255	255	255
255	127	145	200	200	175	175	95	47	255	255
255	127	145	145	175	127	127	95	47	255	255
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255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255

0 = black; 255 = white

Gray Scale Image Analysis



(a) Input



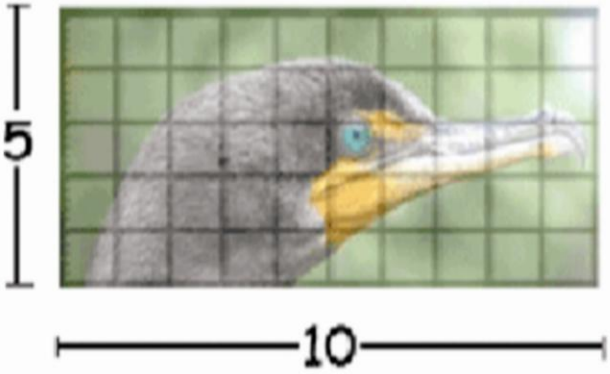
Input patches

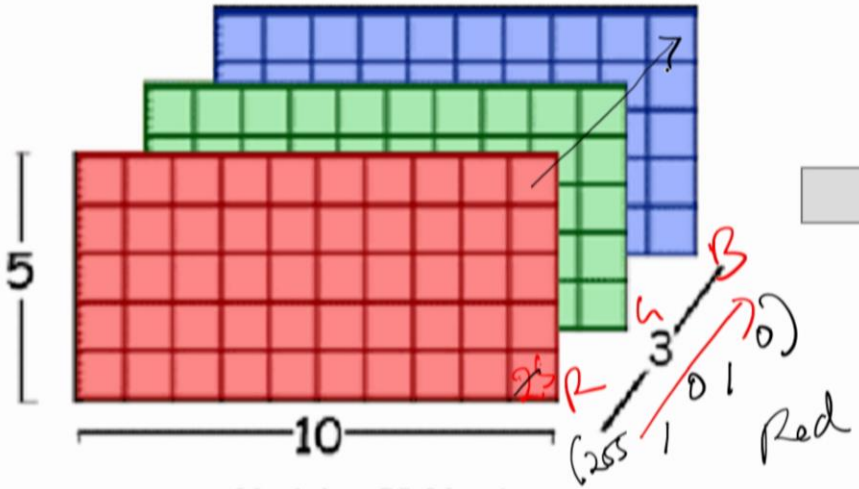
Output patches

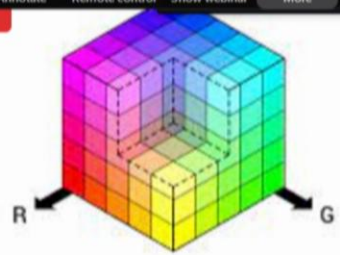
Noise-aware Detail Enhancement in Scanning Electron Microscope imagery. *KLA-Tencor's Neoterix*, 2018

Color Image (3 dimensional)

Audio Video Participants 332 Chat Share Pause Layout Annotate Remote control Show webinar More
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Adarsh Udupi +3 others raised hands View







Color name	RGB triplet	Color
Red	(255, 0, 0)	
Lime	(0, 255, 0)	
Blue	(0, 0, 255)	
White	(255, 255, 255)	
Black	(0, 0, 0)	
Gray	(128, 128, 128)	
Fuchsia	(255, 0, 255)	
Yellow	(255, 255, 0)	
Aqua	(0, 255, 255)	
Silver	(192, 192, 192)	
Maroon	(128, 0, 0)	
Olive	(128, 128, 0)	
Green	(0, 128, 0)	
Teal	(0, 128, 128)	
Navy	(0, 0, 128)	
Purple	(128, 0, 128)	

Color Image Analysis

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Adarsh Udupi +3 others raised hands View



(a) I/p (b) Blurred: O/p (c) Deblurred: O/p

(1) Semantic Segmentation



(a) Blurred: O/p (a) Deblurred: O/p

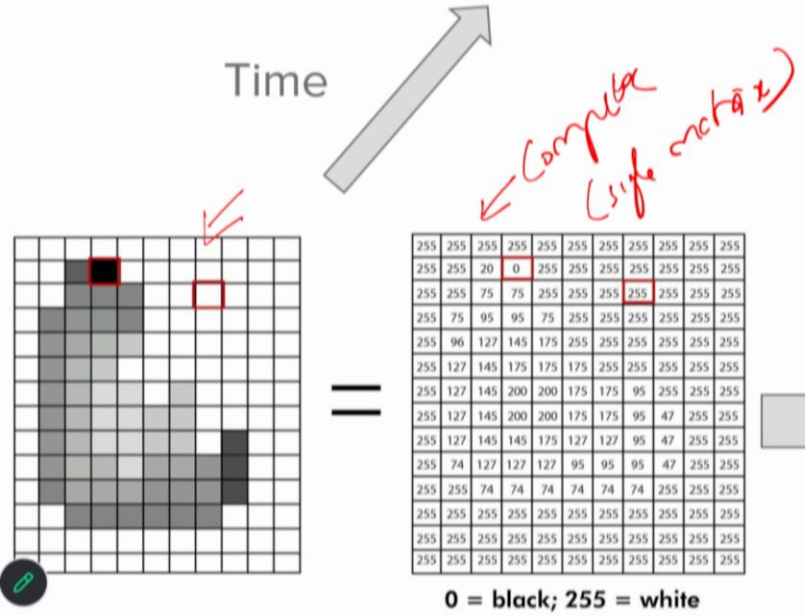
(2) Object Detection

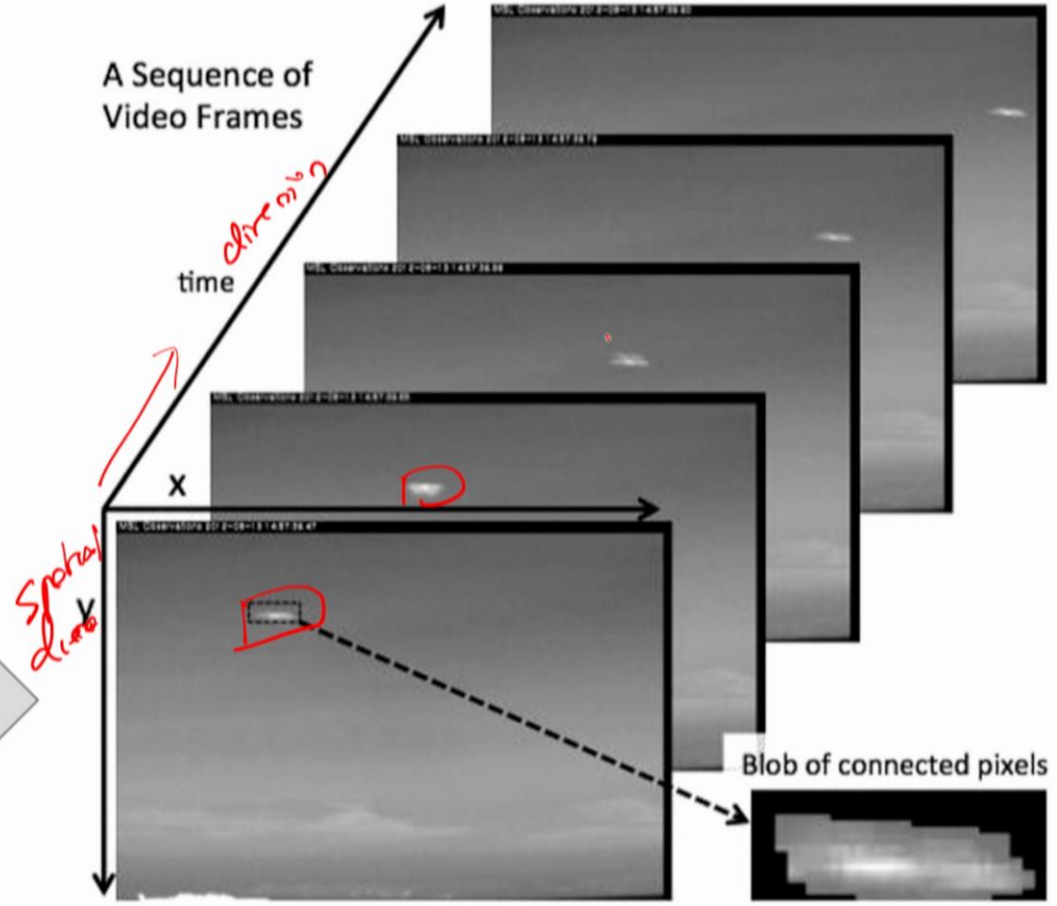
Deep Dynamic Scene Deblurring for Unconstrained Dual-lens Cameras, *IEEE Transactions on Image Processing*, 2021

Grayscale Video Data (3 dimensional)

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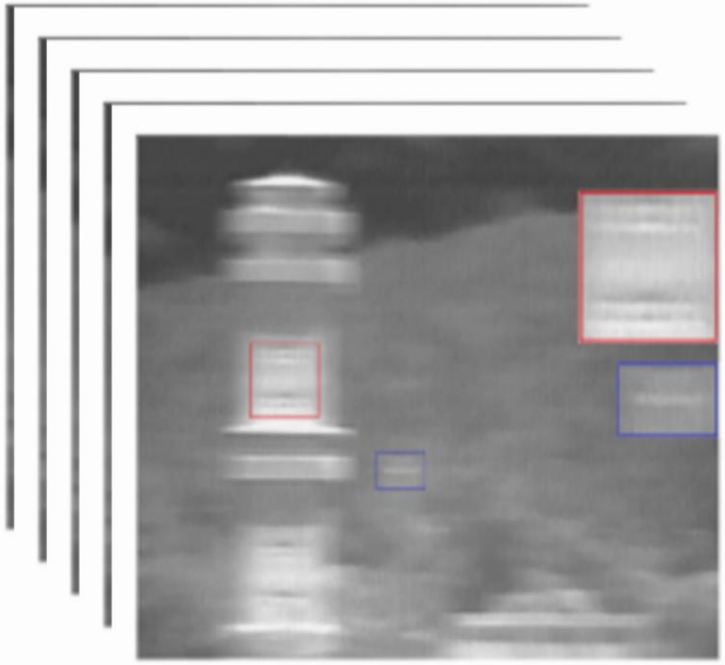




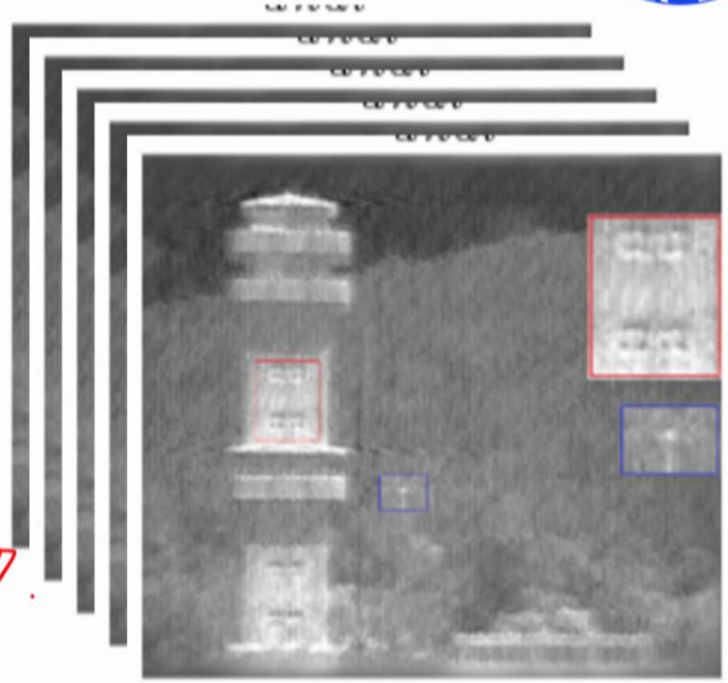


Grayscale Video Analysis

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Adarsh Udupi +3 others raised hands View



Blurred



3(c) k_{anal}+WF

00:20:11:14 01:26:33:03

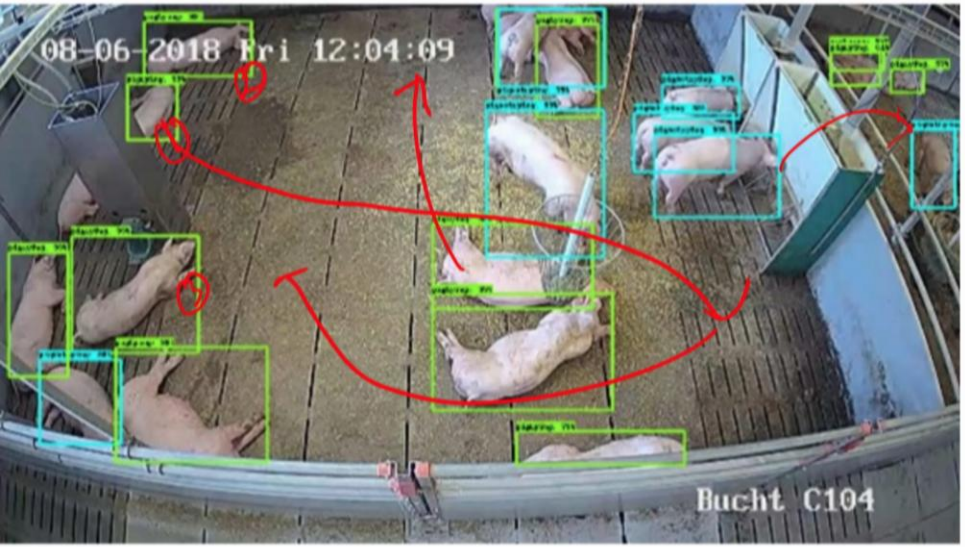
Fast Motion Deblurring of IR Images, *IEEE Signal Processing Letters*, 2022

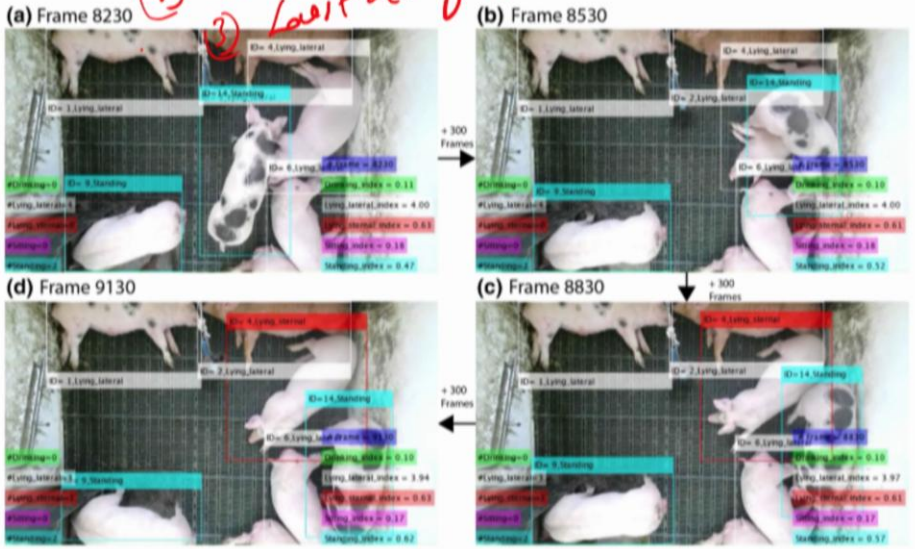
Audio Video Participants 348 Chat Share Pause Layout Annotate Remote control Show webinar More
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Adarsh Udupi +3 others raised hands View

Color Video Analysis

ILLINOIS URBANA-CHAMPAIGN

① Identify each pig
② Actually recognize
③ Count actually work..





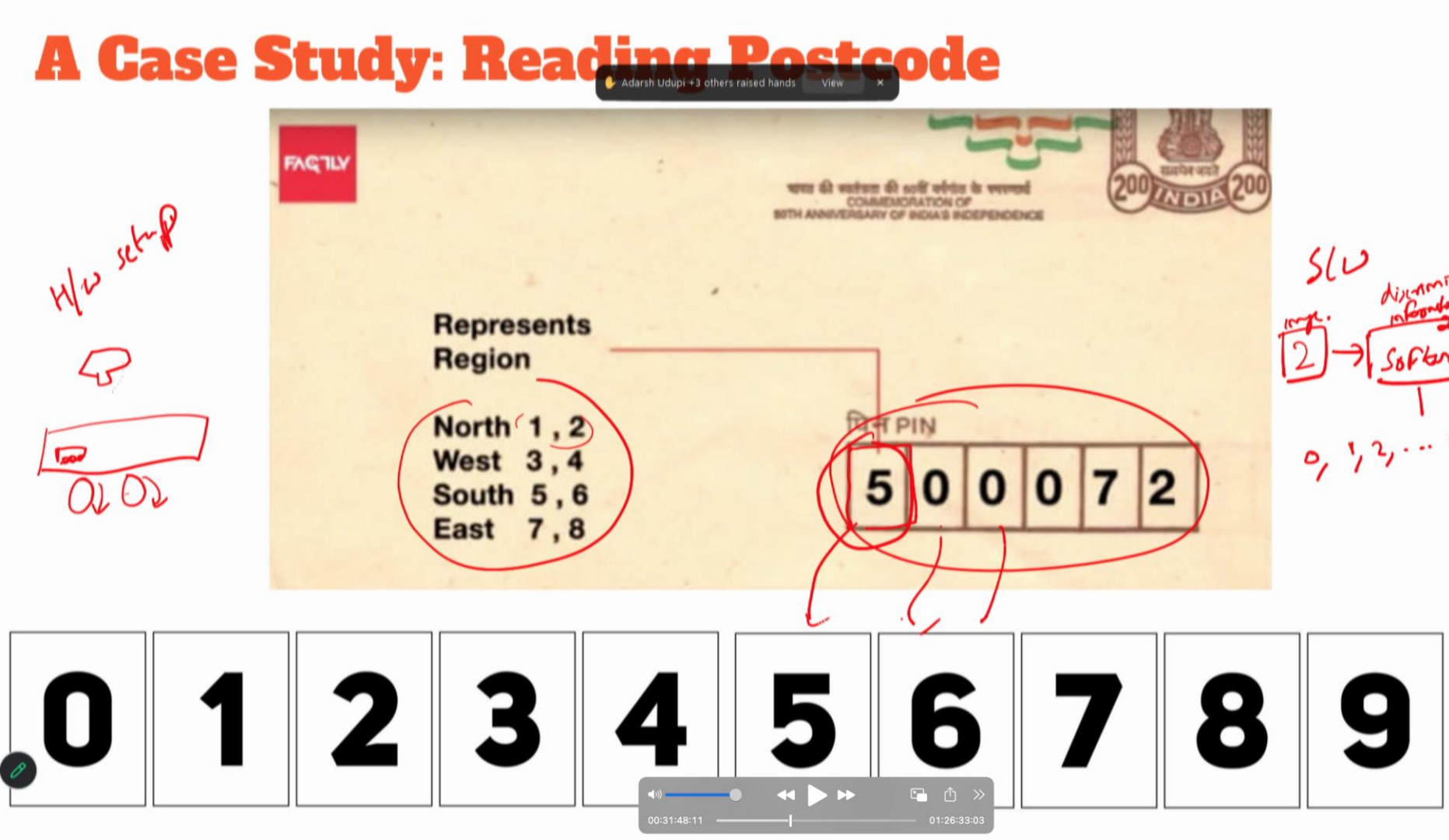
Activity

00:22:50:20 01:26:33:03

Barriers to Computer Vision Applications in Pig Production Facilities, *Computers and Electronics in Agriculture*, 2022

A Case Study: Reading Postcode

Adarsh Udupi +3 others raised hands View



00:31:48:11 01:26:33:03

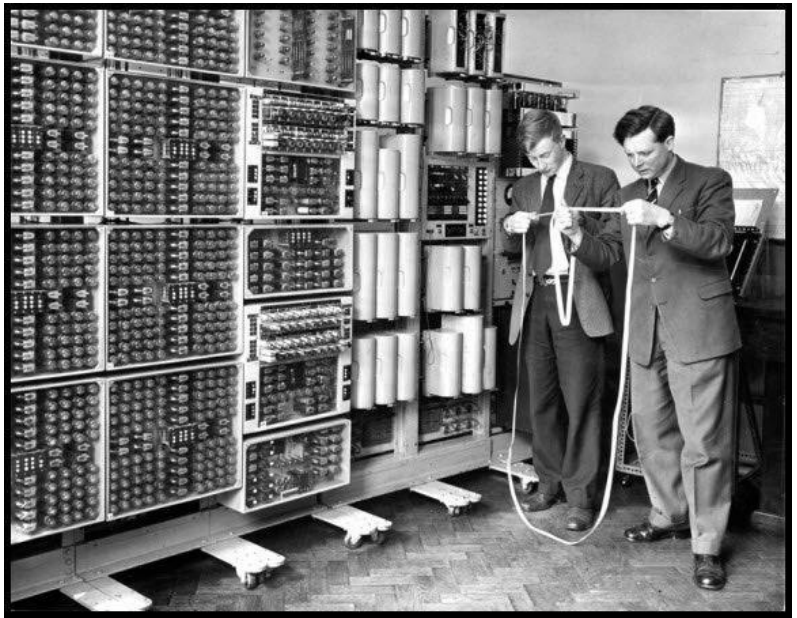
Why ML didn't take-off in 90's?

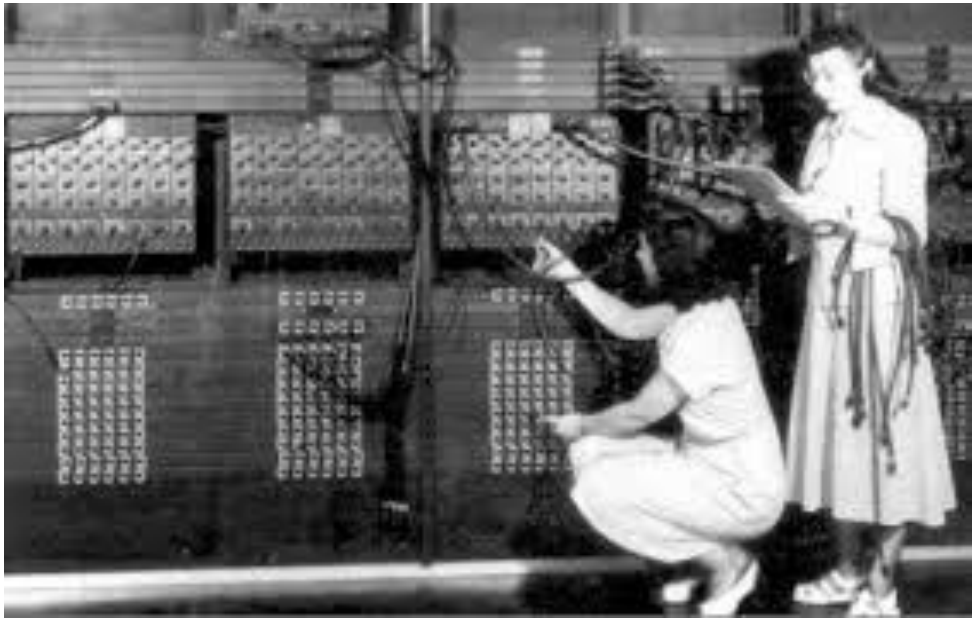
Most techniques for ML are already developed before 1990s!

Why ML didn't take-off in 90's?

Most techniques for AI are already developed before 1990s!

1. ***Limited*** *big data availability*
2. ***Limited*** *computational power to crunch data*





Why ML is trending now?

Big data availability

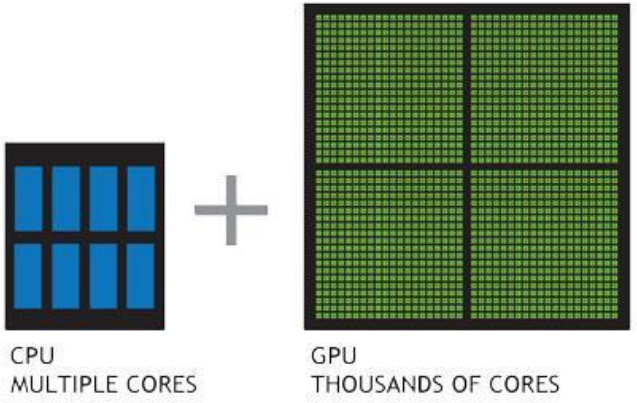
One trillion images.

350 million images uploaded **per day**.

100 hrs of video uploaded **per minute**.

2.5 Petabytes data **every minute**.

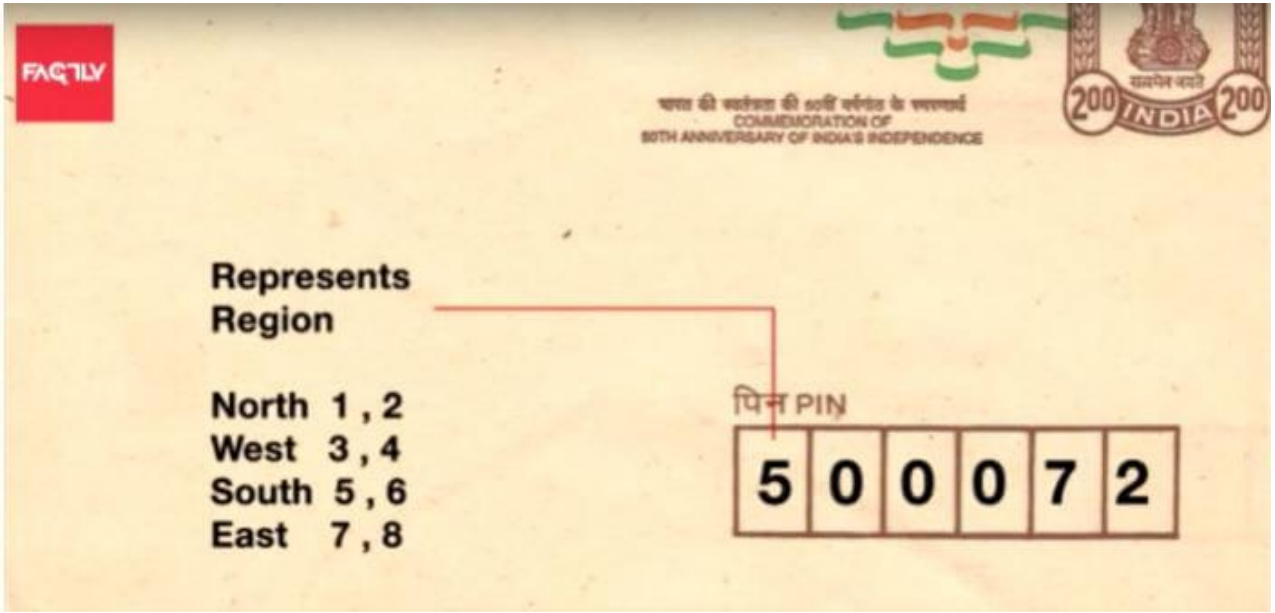
Computational power to crunch data





Parallel processing units - GPUs

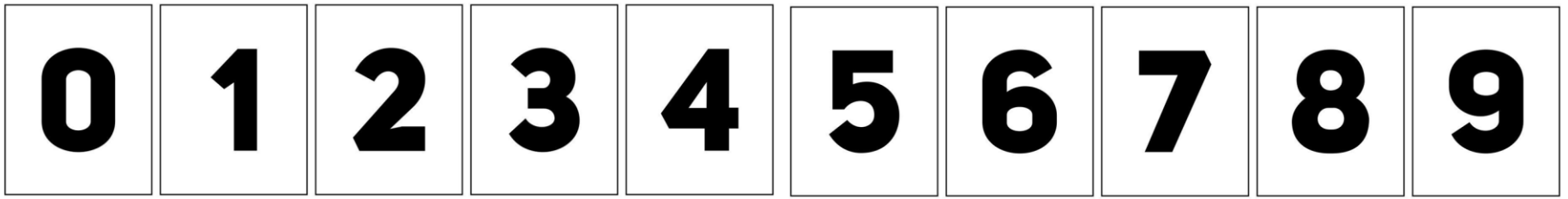
A Case Study: Reading Postcode



0	1	2	3	4	5	6	7	8	9

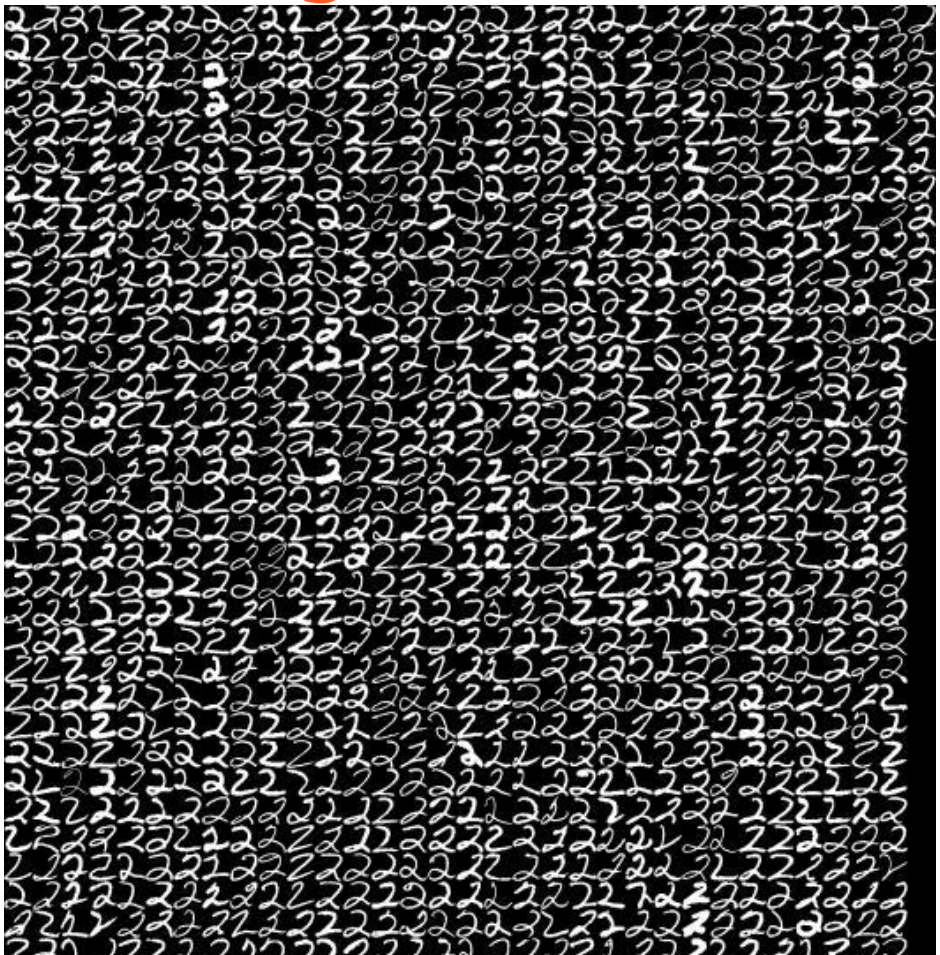
A Case Study: Reading Postcode



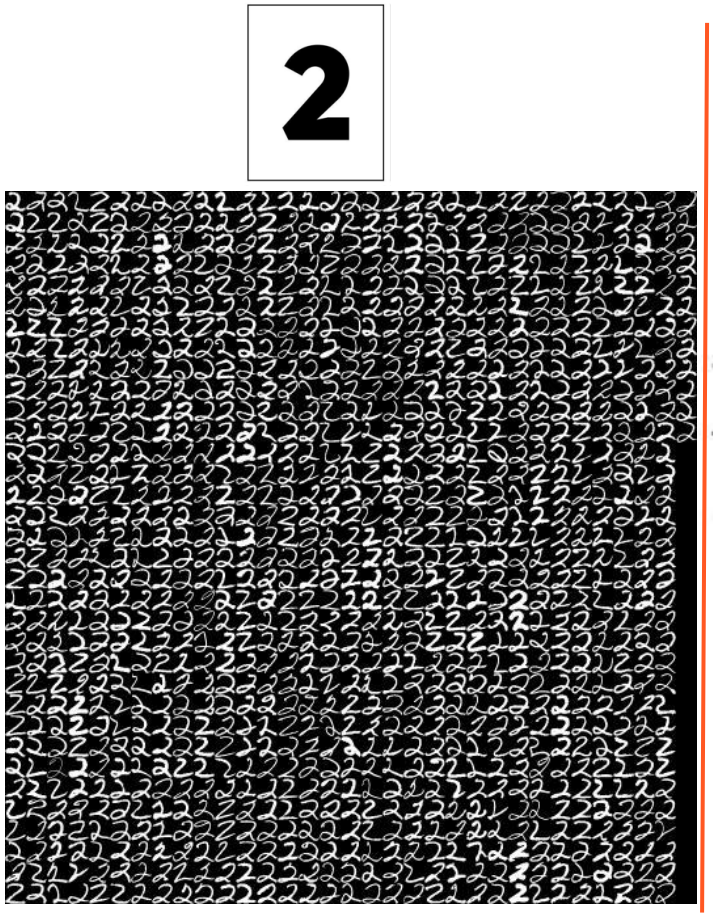


A Case Study: Reading Postcode





A Case Study: Reading Postcode



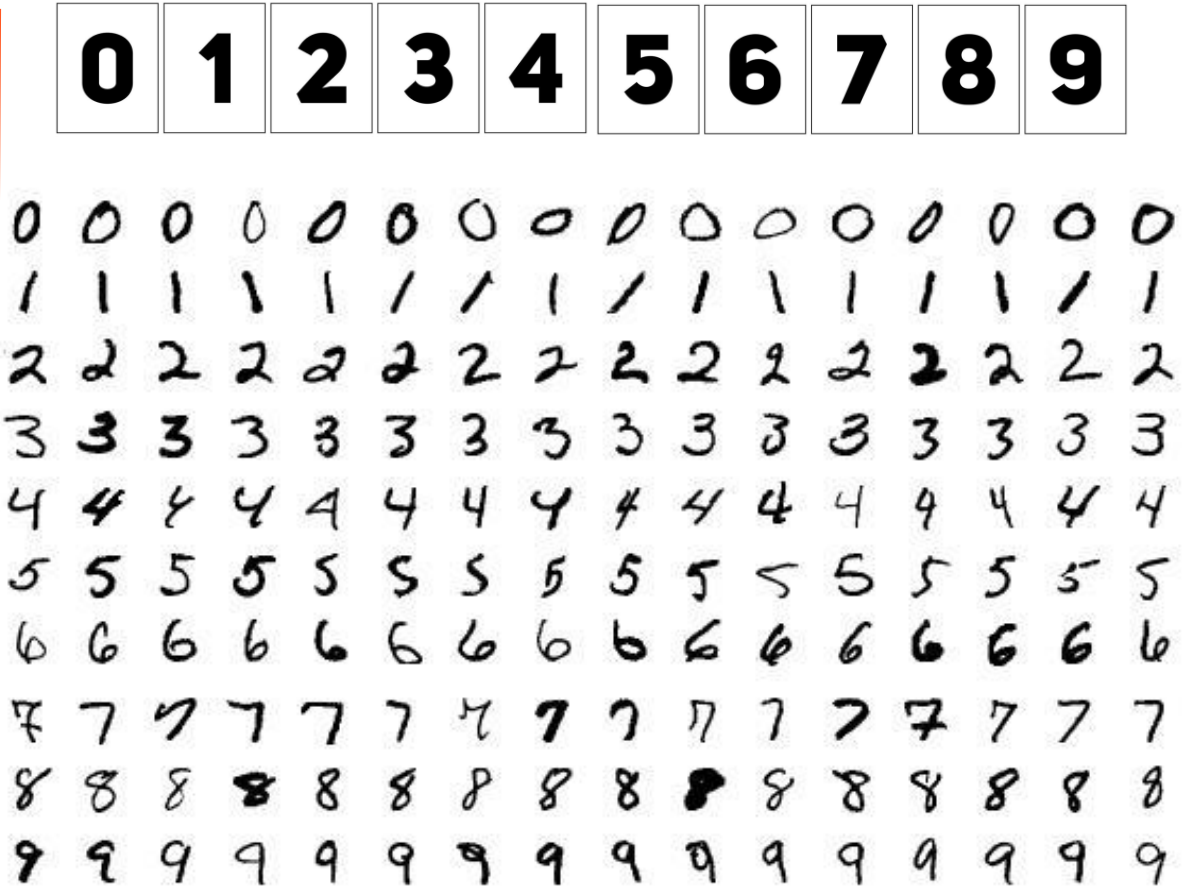
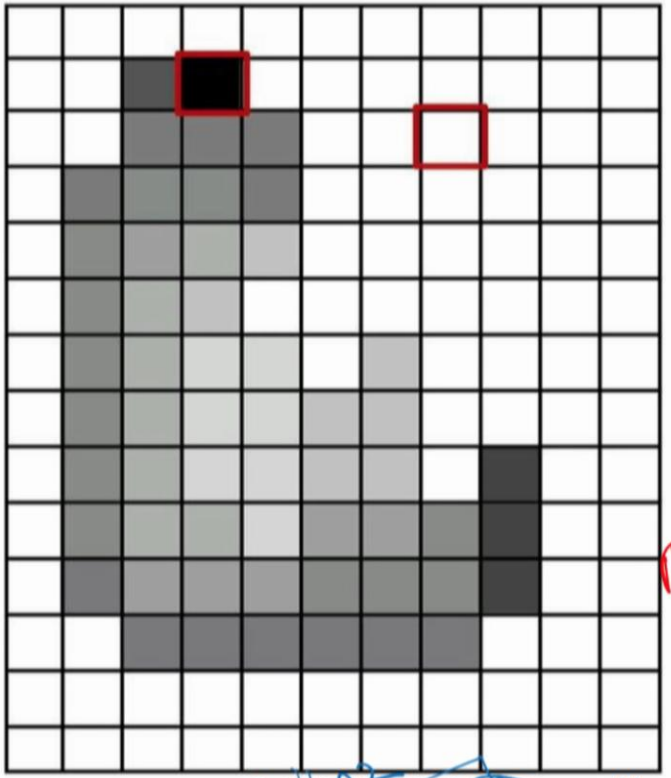


Image Representation in Computer

Adarsh Udupi +4 others raised hands View

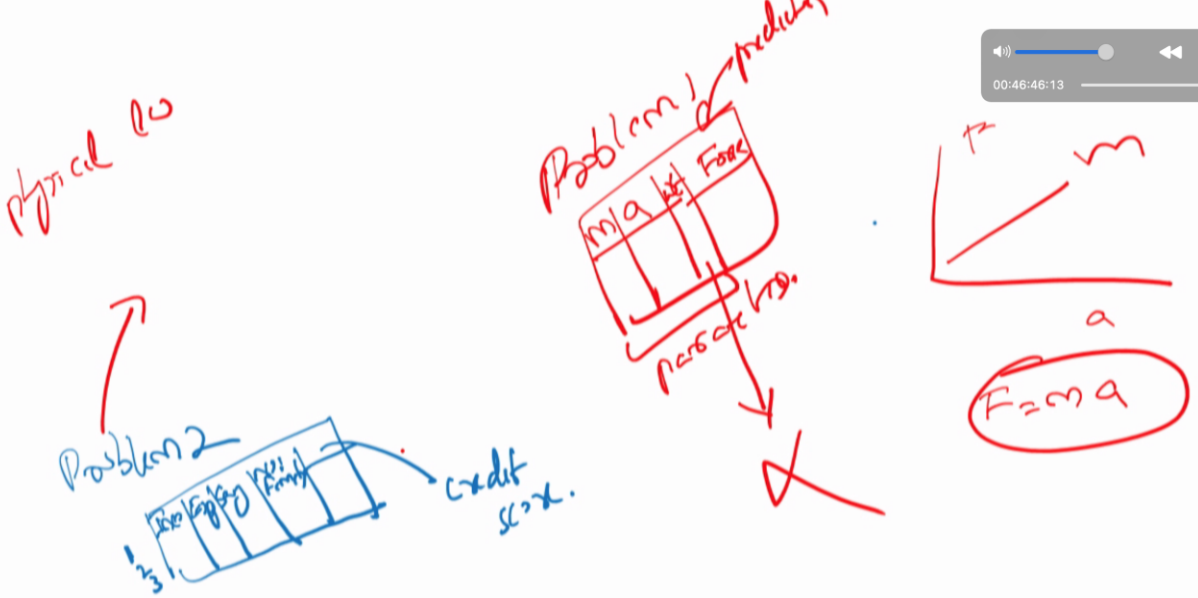
00:42:53:09 01:26:33:03



=

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255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255

0 = black; 255 = white



Sumanta Gupta to Hosts and panelists 10:43

1. AI, 2. Not AI (deterministic)

Dipanjan Biswas to Hosts and panelists 10:43

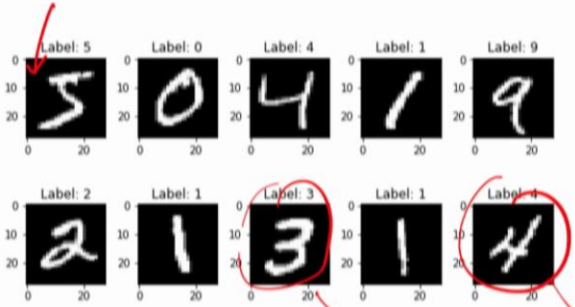
Prob 1: Deterministic (rule based- not AI)

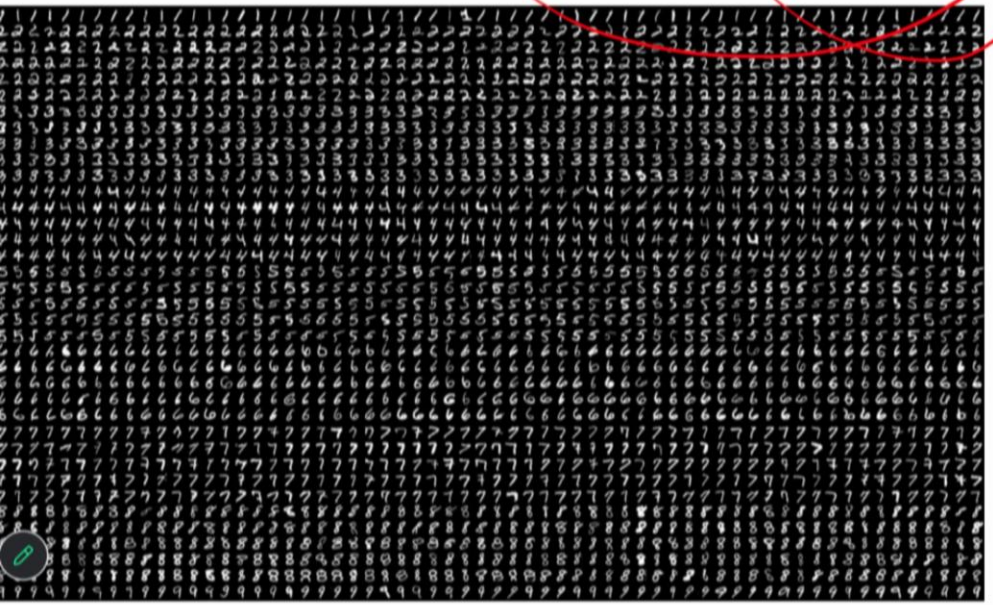
Prob 2: AI

You are screen sharing Stop share

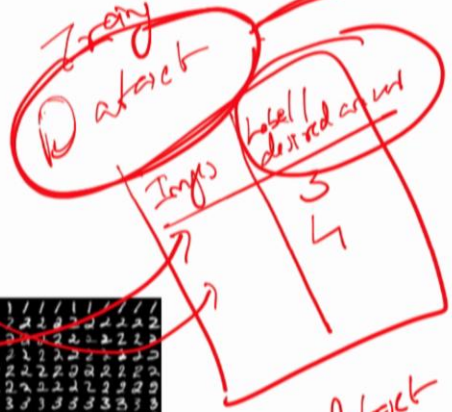
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AI Based Solution: Reading Postcode



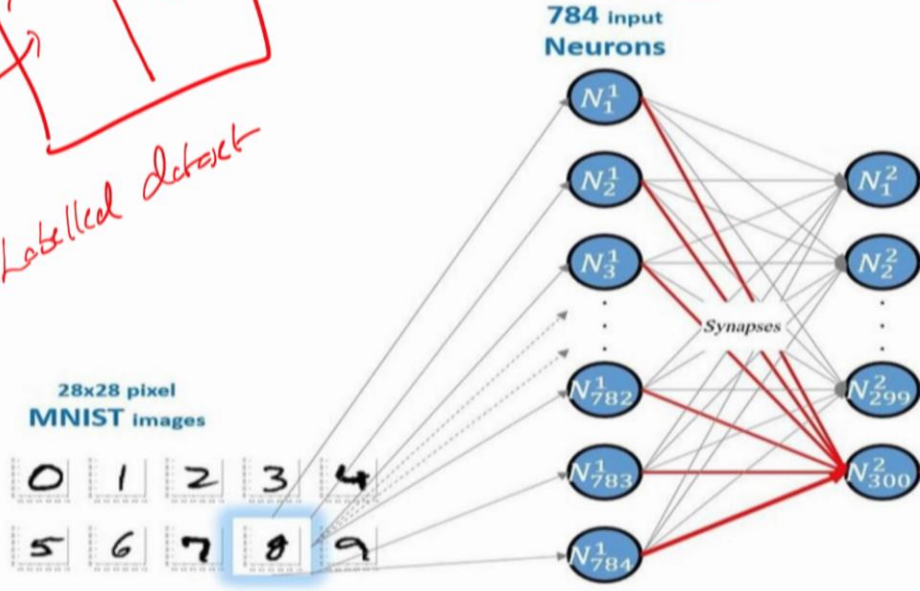


Data and Labels



Labelled dataset

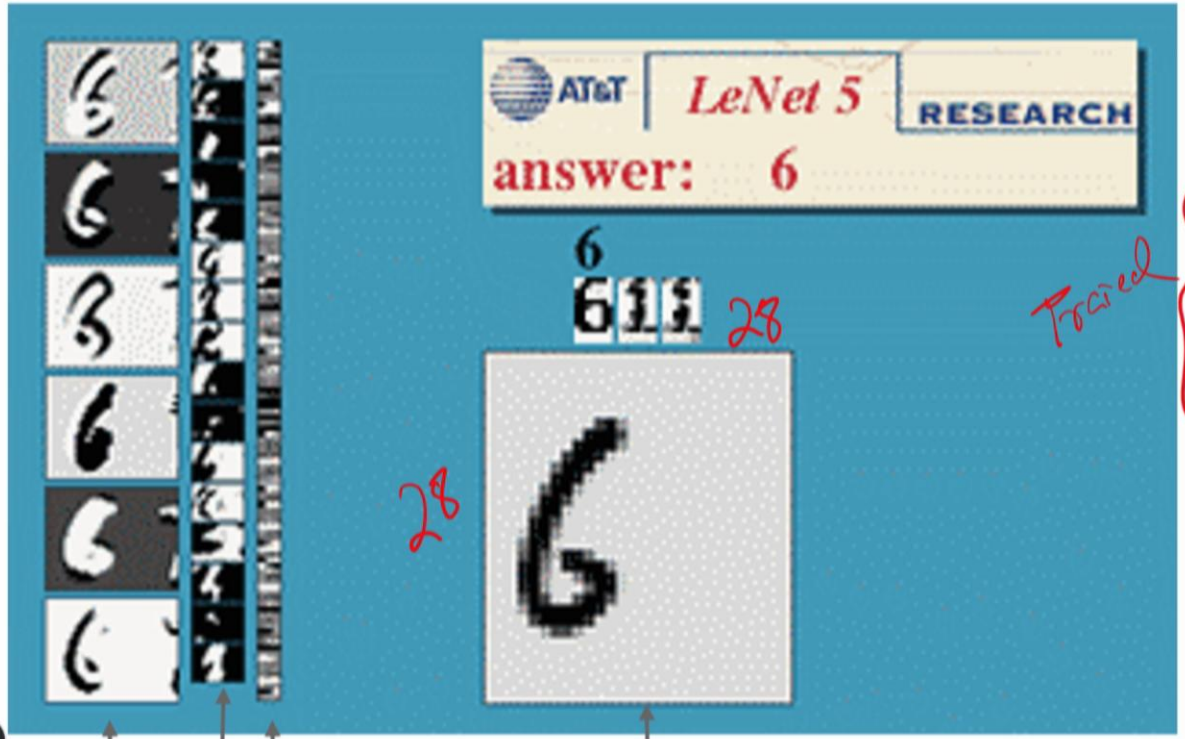




vork and Training

AI for Reading Postcode [LeCun *et al.*].

10 class



Trained

All images are 28x28 grayscale.

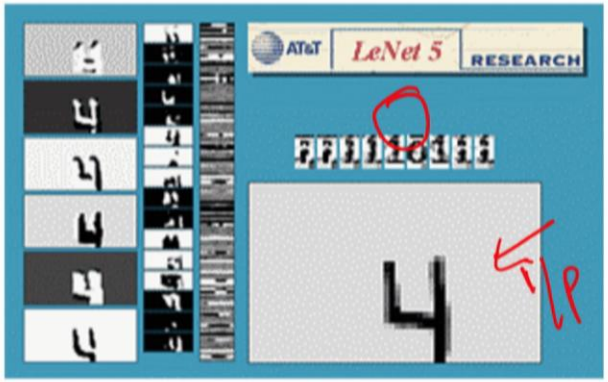
60k training examples.

10k test examples → 1:D

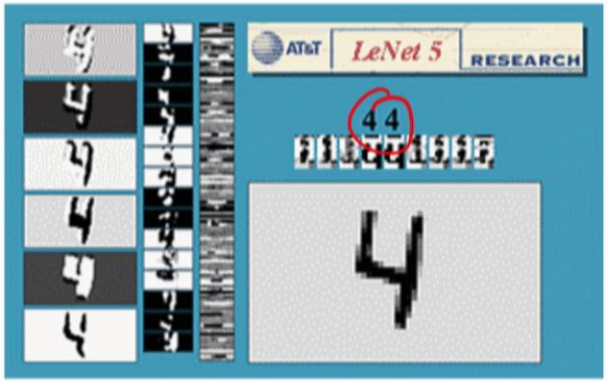
Output value is integer from 0-9

00:52:01:11 01:26:33:03

AI for Reading Postcode [LeCun *et al.*].



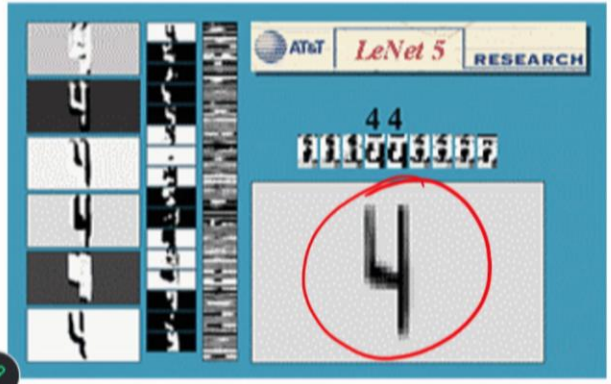
Translation invariance



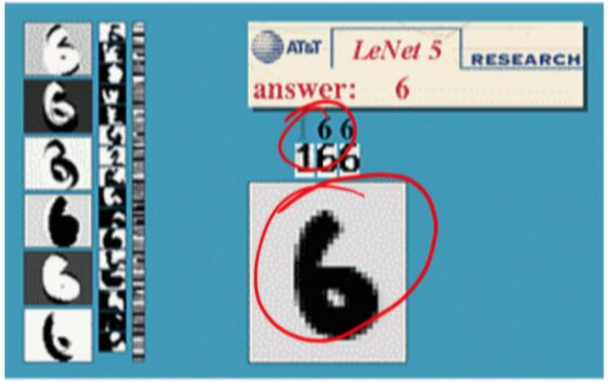
Rotation invariance



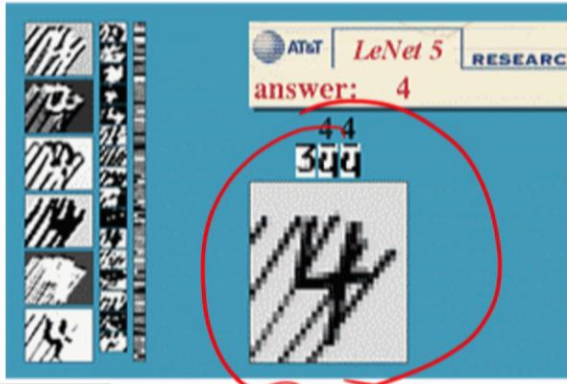
Scale invariance



Squeeze invariance



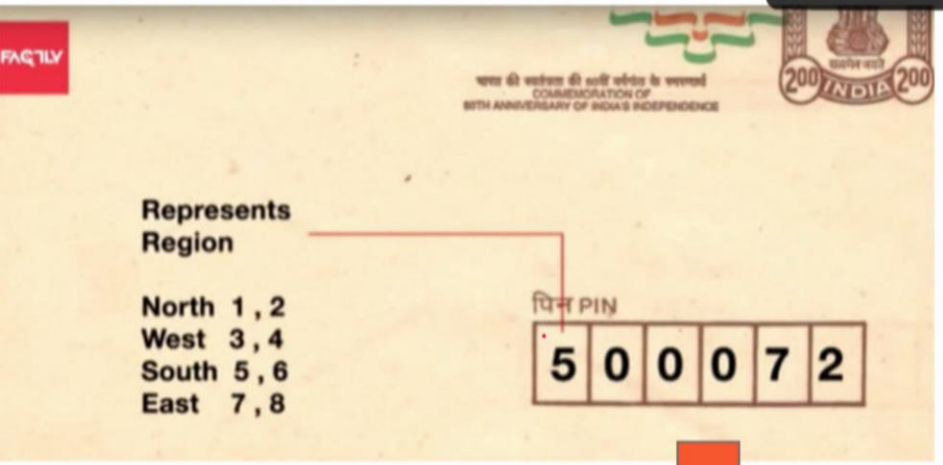
Stroke-width



Noise invariance

AI: Inference/Testing

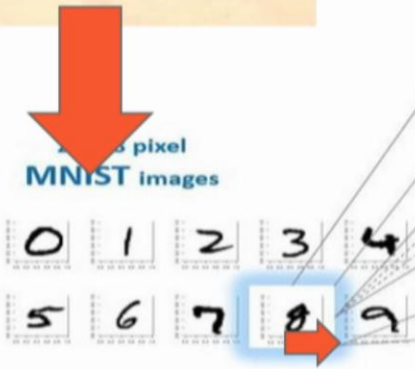
Adarsh Udupi +5 others raised hands View

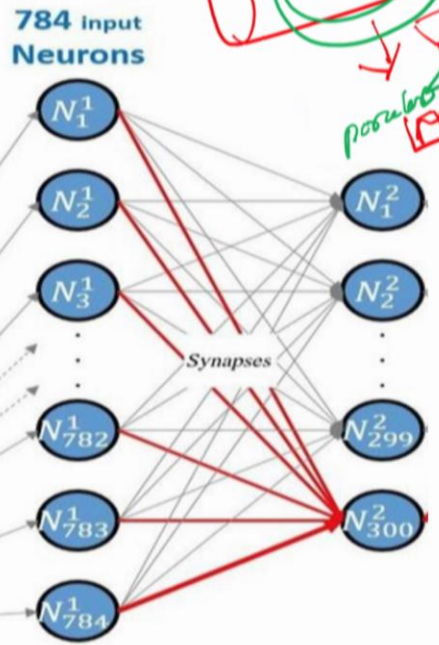


Data Mining

Knowledge distillation

Pattern recognition







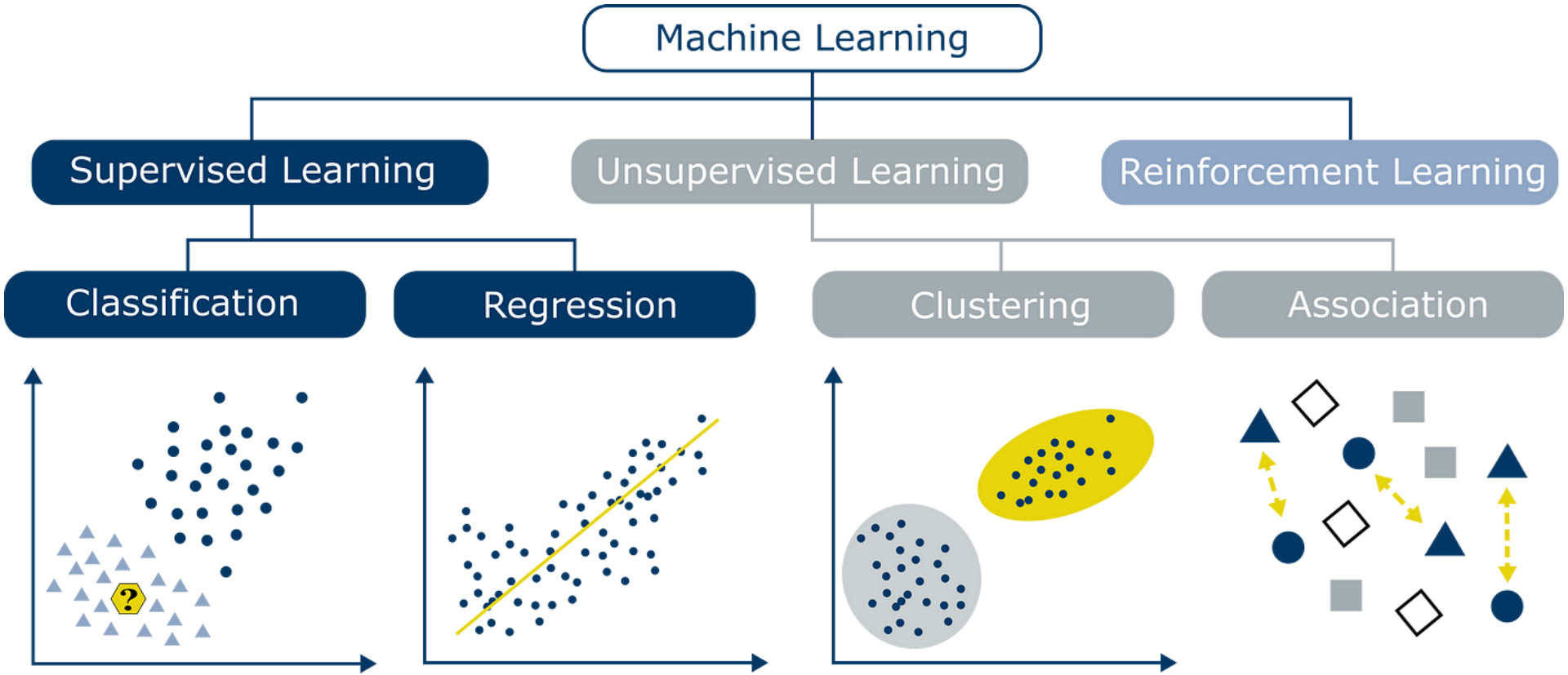
Testing samples extracted from envelope

00:56:35:04 01:26:33:03

etwork (ANN)

O/p

Outline


Machine Learning
Supervised Learning
Unsupervised Learning
Reinforcement Learning
Classification
Regression
Clustering
Association
?

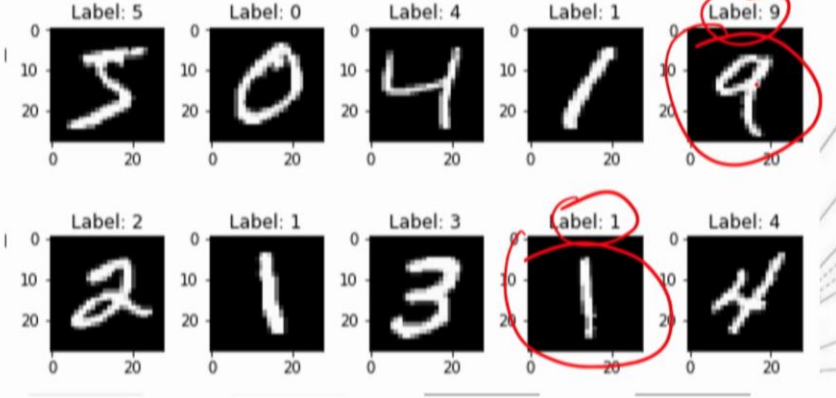
Introduction to Supervised Learning


Adarsh Udupi +5 others raised hands
View
Learning from examples
Yellow color
00:59:59:19
01:26:33:03

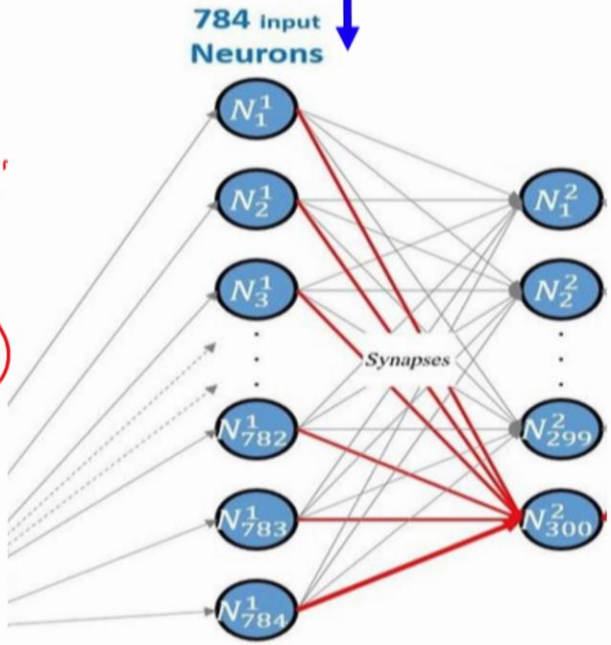
AI: How to Solve it?

Adarsh Udupi +5 others raised hands View ×

1. Collect Labelled Dataset
2. Design Machine Learning Model
3. Define Loss Function
4. Optimize weights

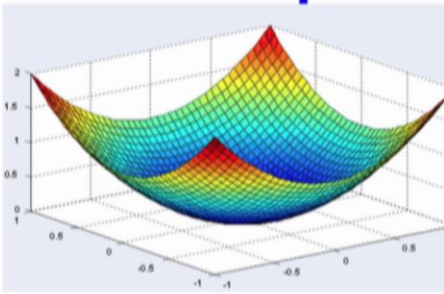


1. Collect Labelled dataset



2. Design Artificial Neural Network

4. Optimize Weight



3. Loss Function

01:02:29:06 01:26:33:03

Collect Labelled Dataset

Adarsh Udupi +5 others raised hands View ×

01:02:41:06 01:26:33:03

kaggle

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Nurse Stress Prediction Wearable Sensors
Priyank Raval · Updated 6 days ago
Usability **9.4** · 80 MB
1 File (CSV)

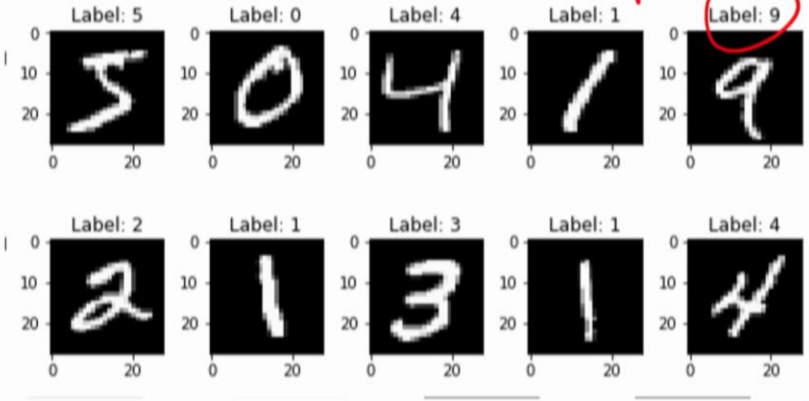
AI: How to Solve it?

Adarsh Udupi +5 others raised hands View

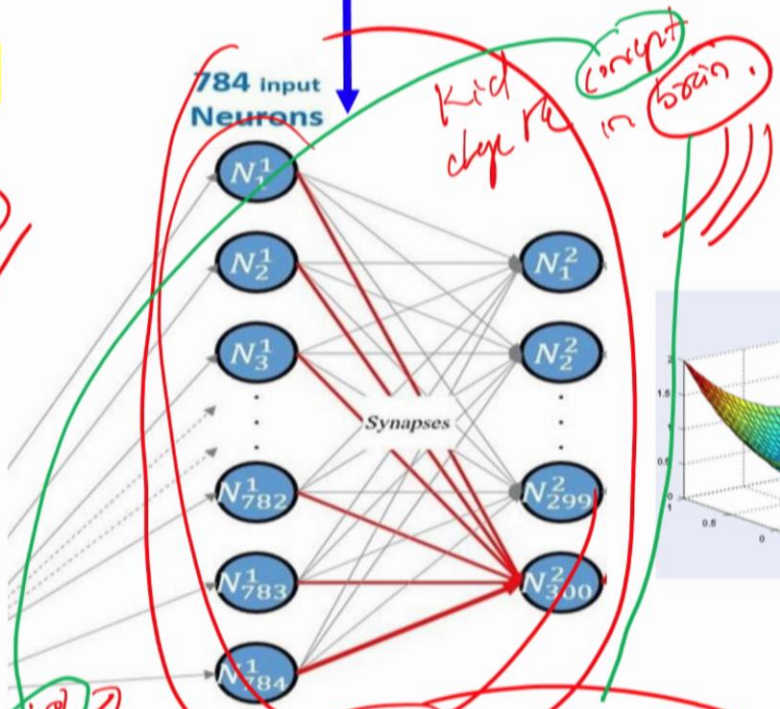
01:08:31:15 01:26:33:03

4. Optimize Weight

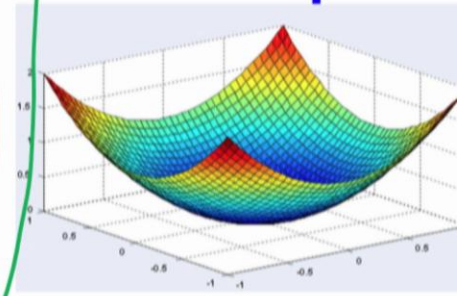
1. Collect Labelled Dataset
2. Design Machine Learning Model
3. Define Loss Function
4. Optimize weights



1. Collect Labelled dataset



2. Design Artificial Neural Network

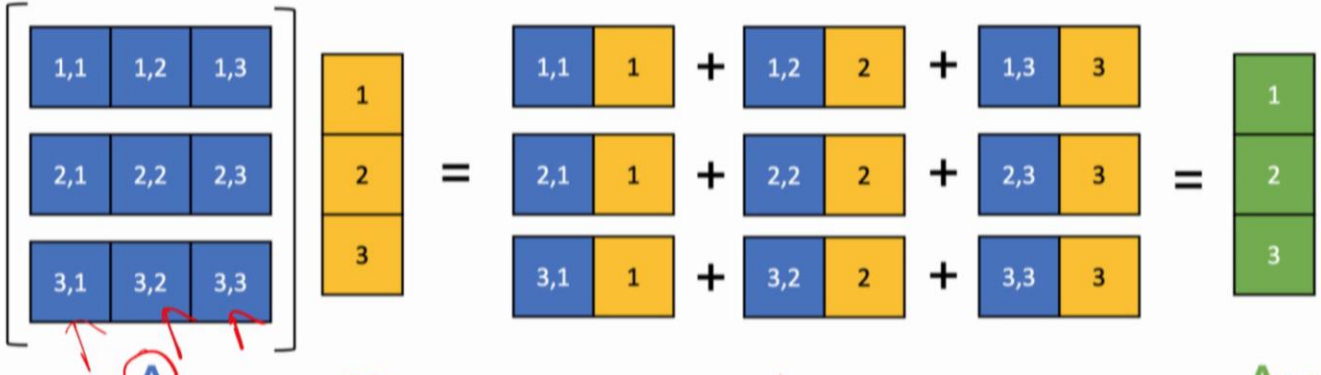


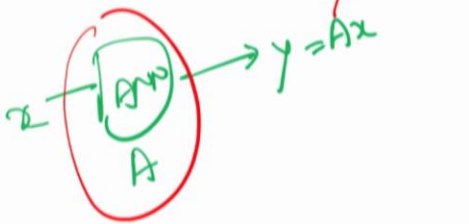
3. Loss Function

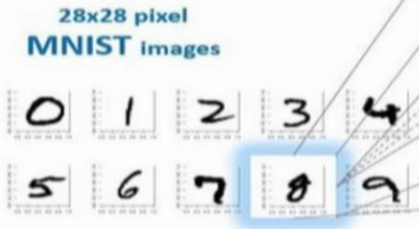
Design ANN Architecture

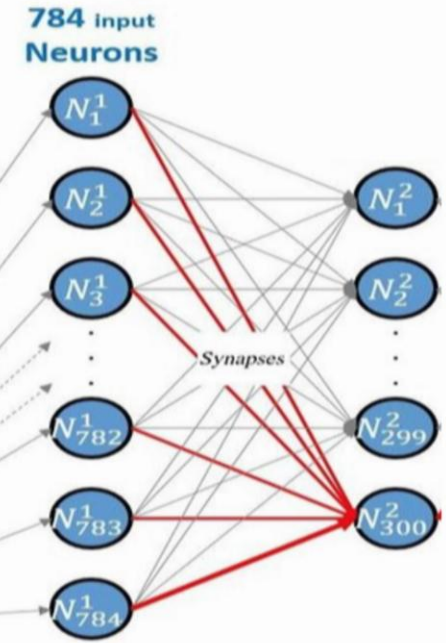
Adarsh Udupi +5 others raised hands View

01:10:03:20 01:26:33:03





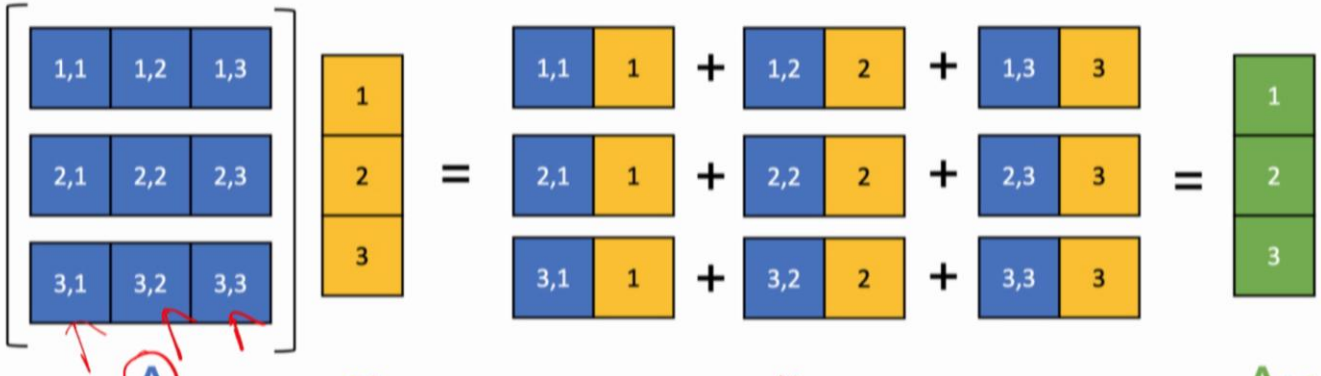


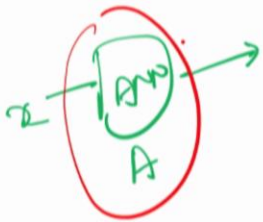


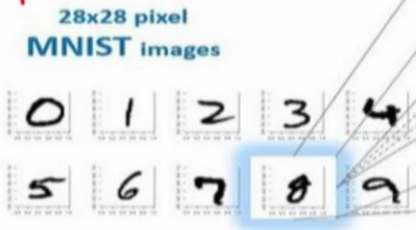
Design ANN Architecture

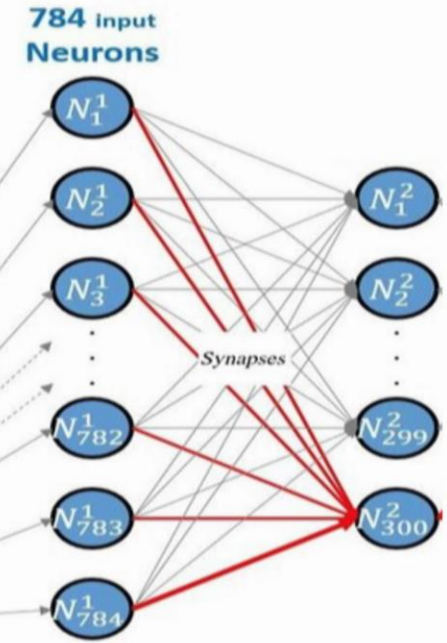
Adarsh Udupi +5 others raised hands View

01:11:16:14 01:26:33:03



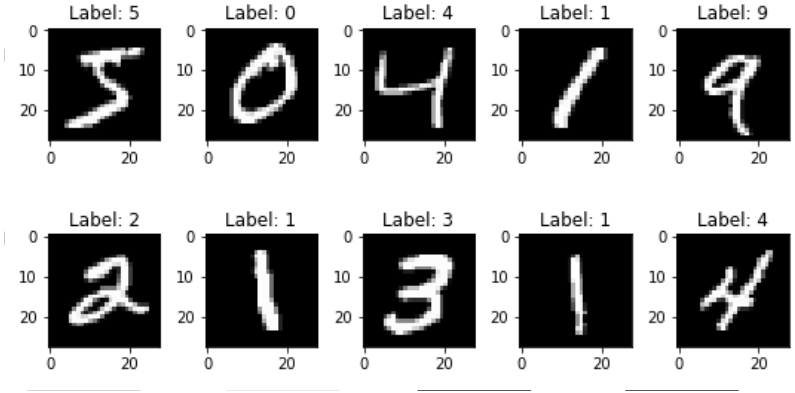






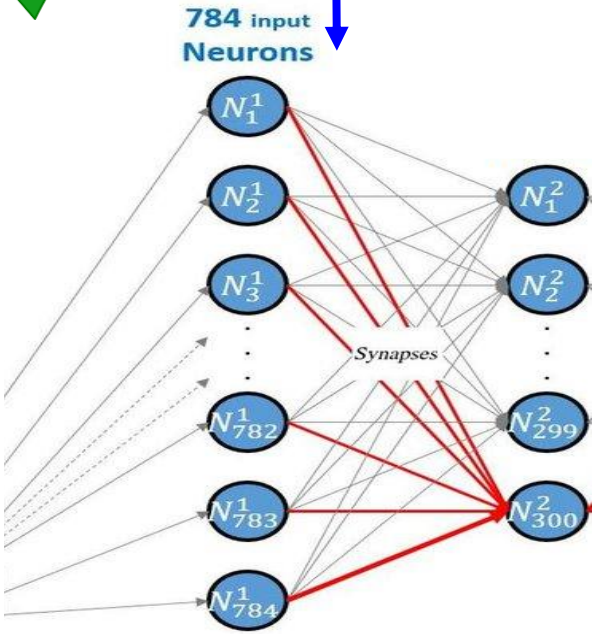
AI: How to Solve it?

1. Collect Labelled Dataset ✓
2. Design Machine Learning Model ✓
3. Define Loss Function
4. Optimize weights

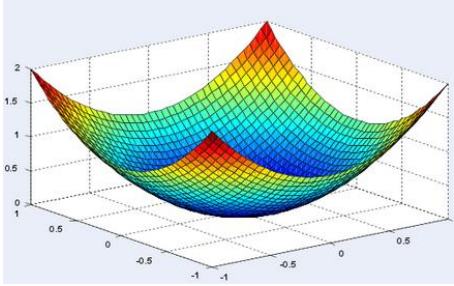


1. Collect Labelled dataset

4. Optimize Weight



2. Design Artificial Neural Network



3. Loss Function

Cross-entropy Loss

Adarsh Udupi +5 others raised hands View ×

01:13:08:21 01:26:33:03



Input image

NN Layers

Logits (L)

3.2
1.3
0.2
0.8

Softmax

$$S(y)_i = \frac{\exp(y_i)}{\sum_{j=1}^{n} \exp(y_j)}$$

Output probabilities (P)

0.775
0.116
0.039
0.070

Classes

Dog
Cat
Horse
Cheetah

What we want?

S

0.775
0.116
0.039
0.070

$L_{CE}(S,T)$

T

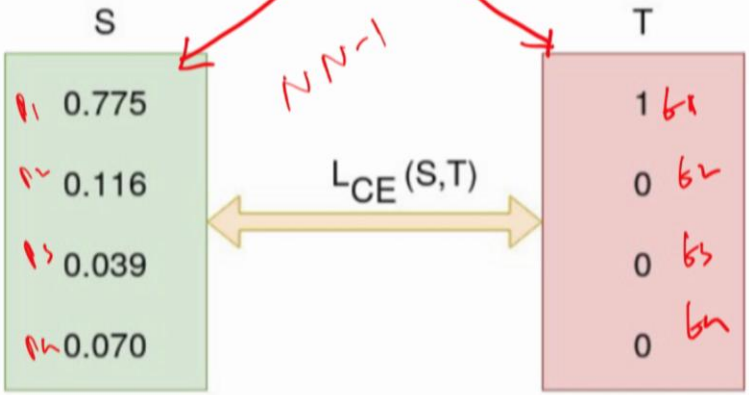
1
0
0
0

Cross-entropy Loss?

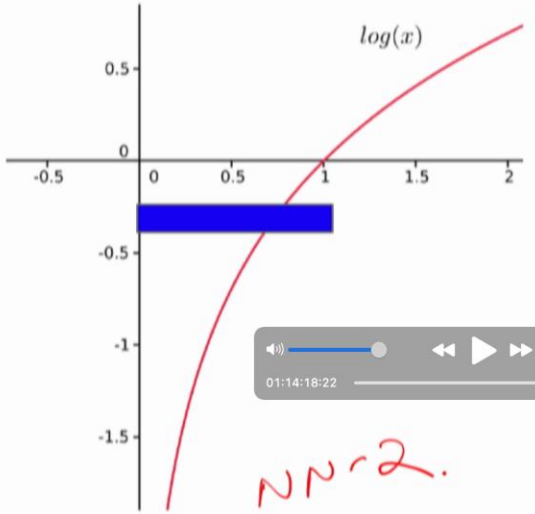
Adarsh Udupi +5 others raised hands View ×

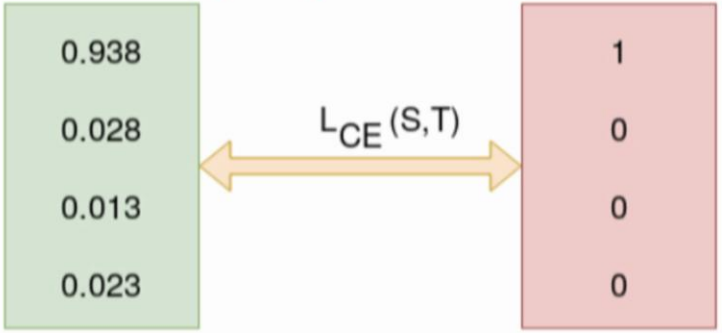
$$L_{\text{CE}} = -\sum_{i=1}^{n} t_i \log(p_i), \quad \text{for n classes,}$$

where t_i is the truth label and p_i is the Softmax probability for the ith class.



$$\begin{aligned} L_{CE} &= -\sum_{i=1} T_i \log(S_i) \\ &= -\left[1\log_2(0.775) + 0\log_2(0.126) + 0\log_2(0.039) + 0\log_2(0.070)\right] \\ &= -\log_2(0.775) \\ &= 0.3677 \end{aligned}$$





$$\begin{aligned} L_{CE} &= -1\log_2(0.936) + 0 + 0 + 0 \\ &= 0.095 \end{aligned}$$

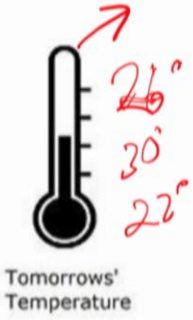
Classification vs Regression

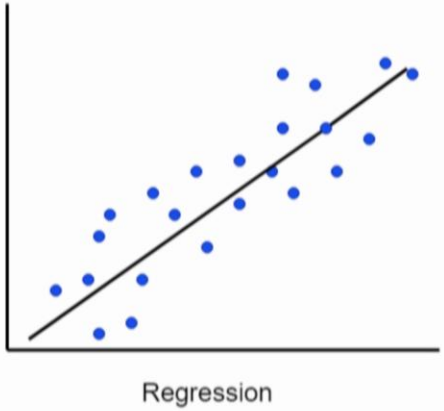
Adarsh Udupi +5 others raised hands View

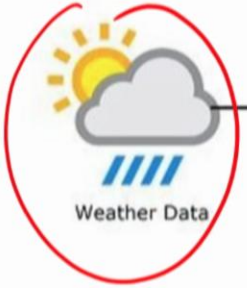


Regression

26.1°
26.1345





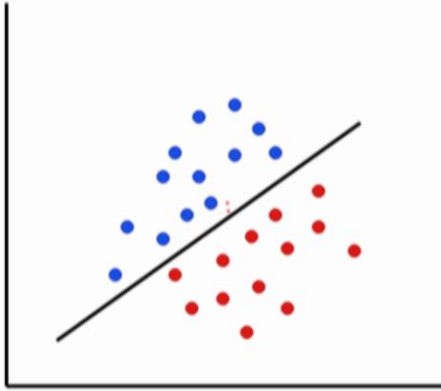


Classification

Hot/Cold



Hot or Cold



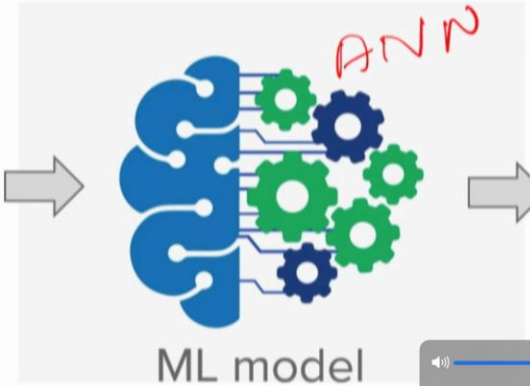
01:17:18:05 01:26:33:03

A Real Life Application















Input and Output are similar

MSE Loss

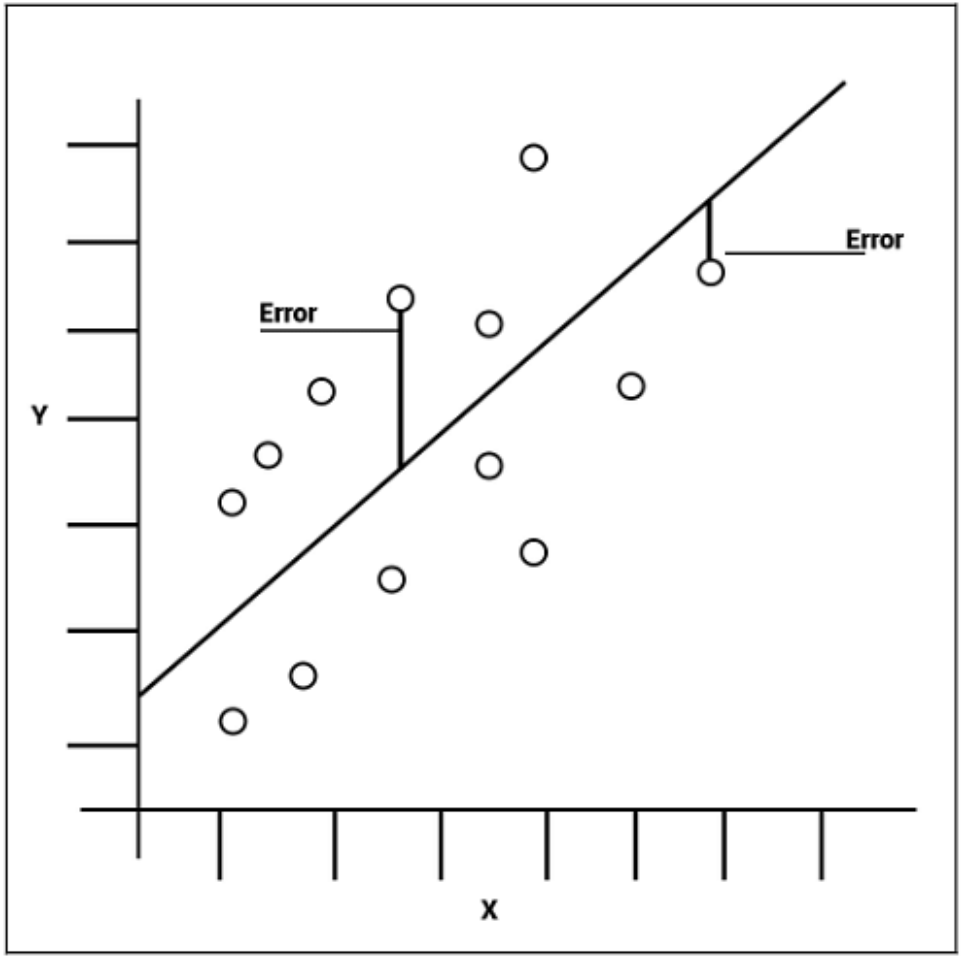
$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

MSE = mean squared error

n = number of data points

Y_i = observed values

$\hat{Y}_i$ = predicted values



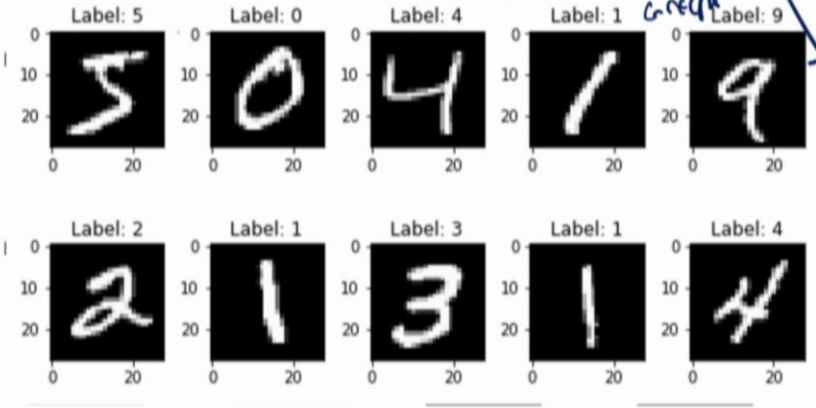
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AI: How to Solve It?

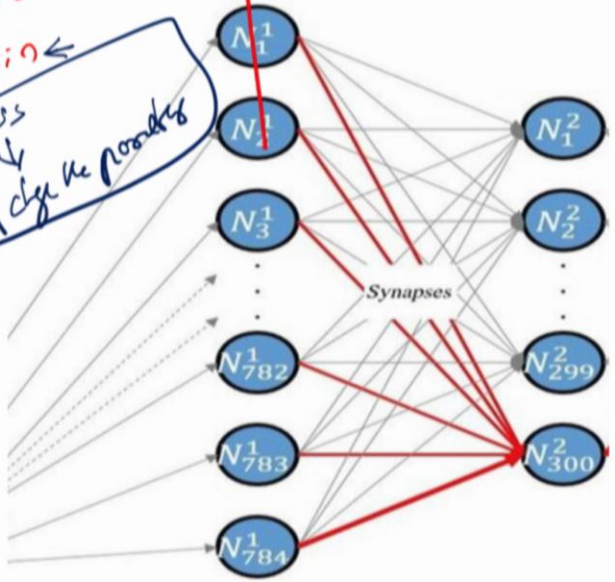
Supervised Learning

1. Collect Labelled Dataset ✓
2. Design Machine Learning Model ✓
3. Define Loss Function ✓
4. Optimize weights

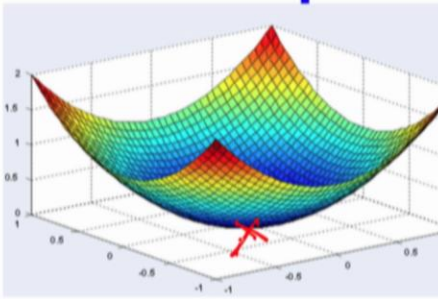
4. Optimize Weight



1. Collect Labelled dataset



2. Design Artificial Neural Network



3. Loss Function

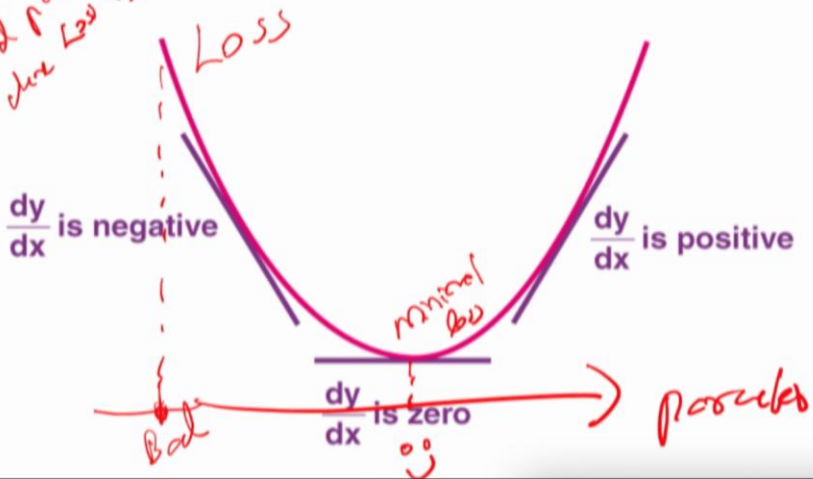
00:11:38:20 02:22:23:01

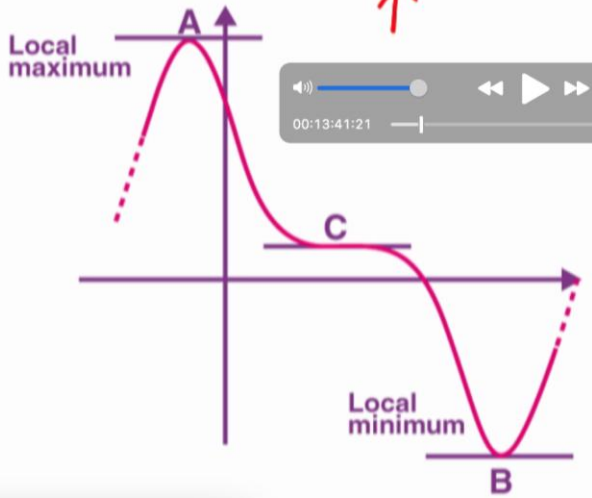
Lecture 02.MOV

Optimizing ANN: Closed Form Expression

	Maximum	Minimum
Necessary condition	$\frac{dy}{dx} = 0$	$\frac{dy}{dx} = 0$
Sufficient condition	$\frac{dy}{dx} = 0$; $\frac{d^2y}{dx^2} < 0$	$\frac{dy}{dx} = 0$; $\frac{d^2y}{dx^2} > 0$

Loss = F(parameters)
objective:
Find parameters
that Loss is minimal





00:13:41:21
02:22:23:01

Gradient descent

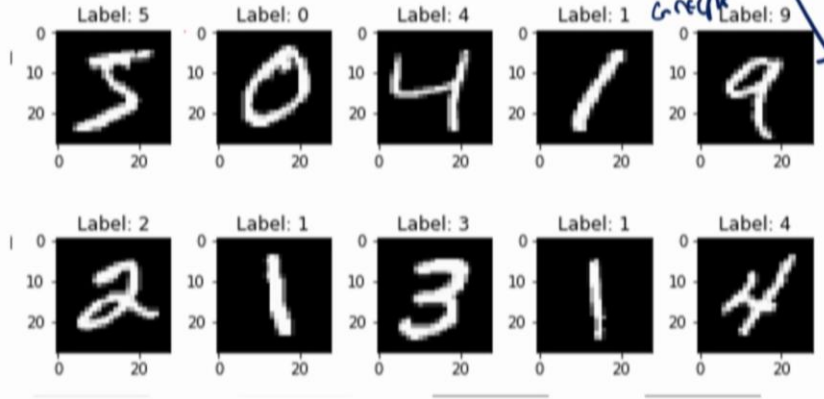




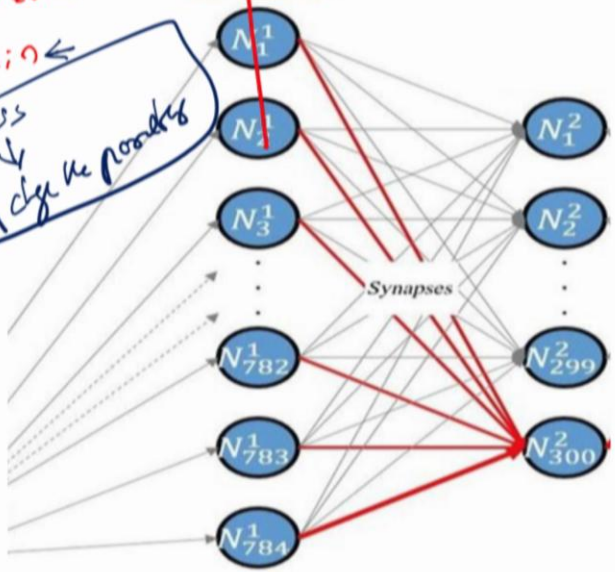
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AI: How to Solve it?

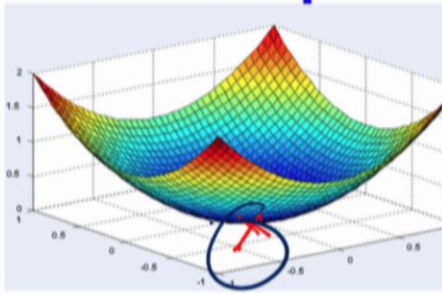
1. Collect Labelled Dataset
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1. Collect Labelled dataset



2. Design Artificial Neural Network



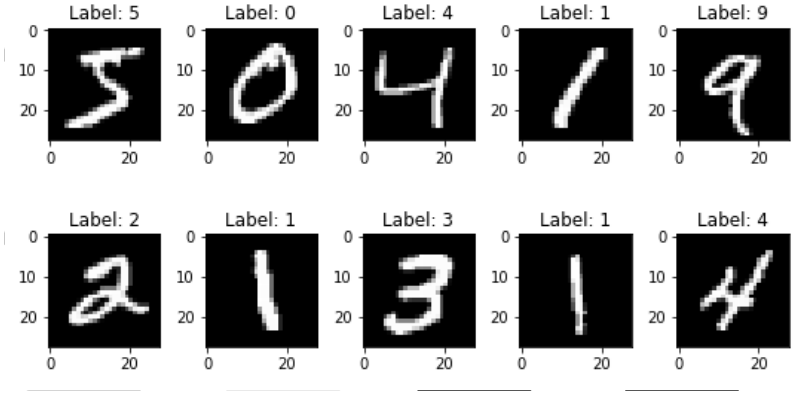
3. Loss Function

4. Optimize Weight

00:19:55:19 02:22:23:01

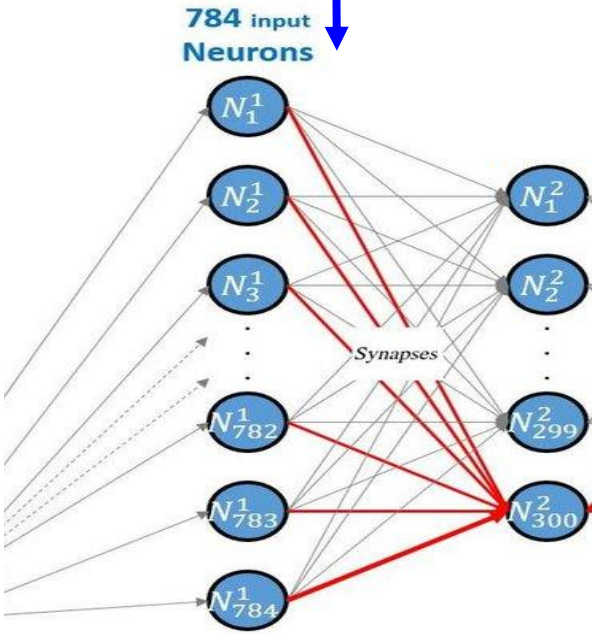
Parametric vs Nonparametric Methods?

1. Collect Labelled Dataset
2. Design ML Model
3. Define Loss Function
4. Optimize weights

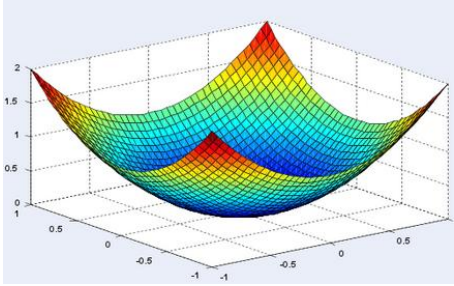


1. Collect Labelled dataset

4. Optimize Weight



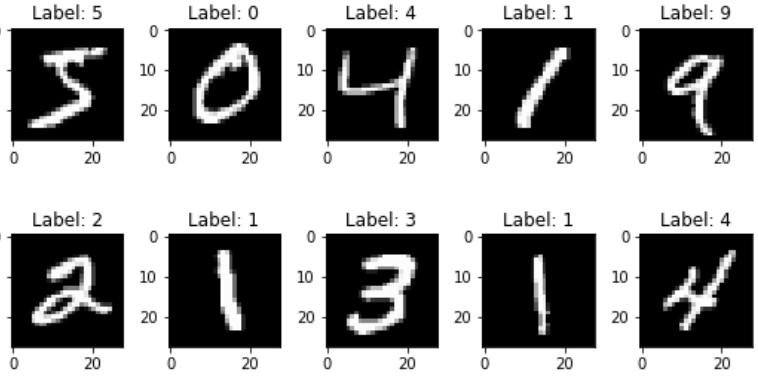
2. Design Artificial Neural Network



3. Loss Function

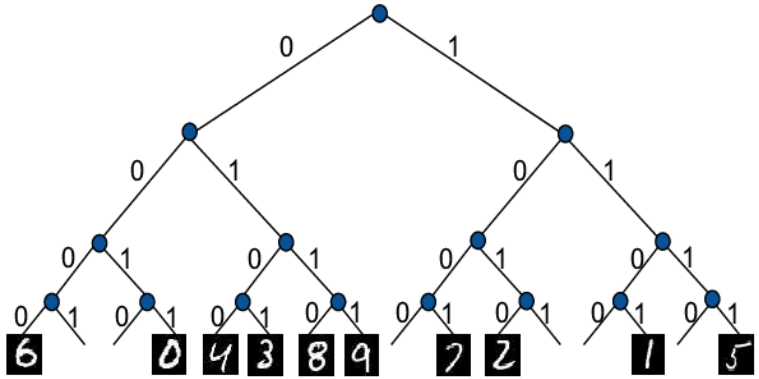
Parametric vs Nonparametric Methods?

1. Collect Labelled Dataset
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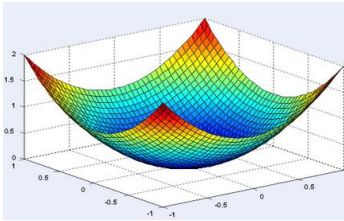


1. Collect Labelled dataset

4. Optimize Weight



2. Design Artificial Neural Network



3. Loss Function



Data Challenge 1: Missing Data

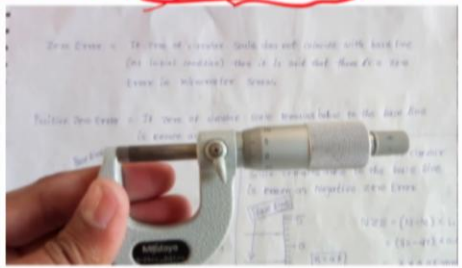
Missing data is a source of error in any data set requiring correction as they can lead to serious problems.

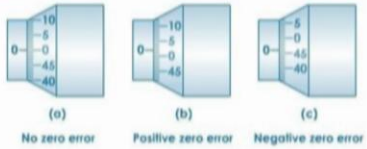
Imputation methods can be used in order to fill those gaps and provide a complete data set.

	School ID	Name	Address	City	Subject	Marks	Rank	Grade
0	101.0	Alice	123 Main St	Los Angeles	Math	85.0	2	B
1	102.0	Bob	456 Oak Ave	New York	English	92.0	1	A
2	103.0	Charlie	789 Pine Ln	Houston	Science	78.0	4	C
3	NaN	David	101 Elm St	Los Angeles	Math	89.0	3	B
4	105.0	Eva	NaN	Miami	History	NaN	8	D
5	106.0	Frank	222 Maple Rd	NaN	Math	95.0	1	A
6	107.0	Grace	444 Cedar Blvd	Houston	Science	80.0	5	C
7	108.0	Henry	555 Birch Dr	New York	English	88.0	3	B

Bin

Data Challenge 2: Errors Vs Artifacts





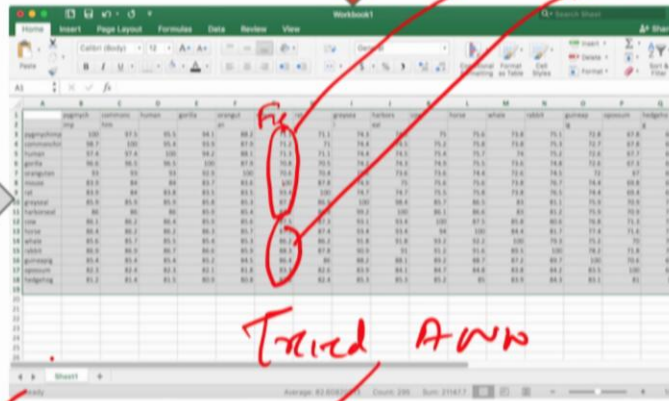
Artifacts



Aspects of Real World



Raw Data



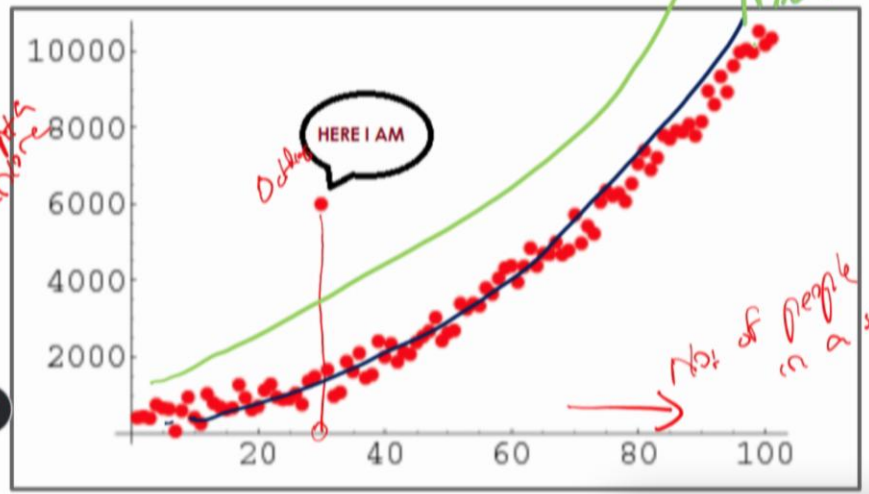
Processed Data

Data Challenge 3: Outliers

Outlier: A data object that **deviates significantly** from the normal objects as if it were **generated by a different mechanism**

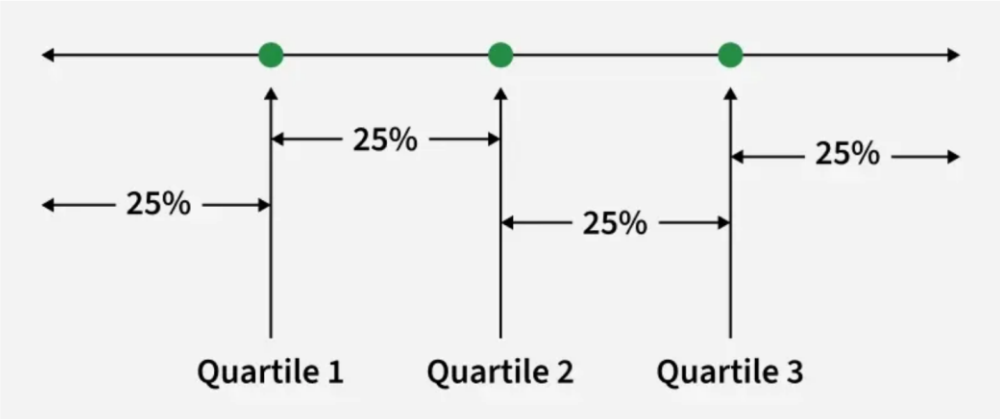
Ex.: Unusual credit card purchase

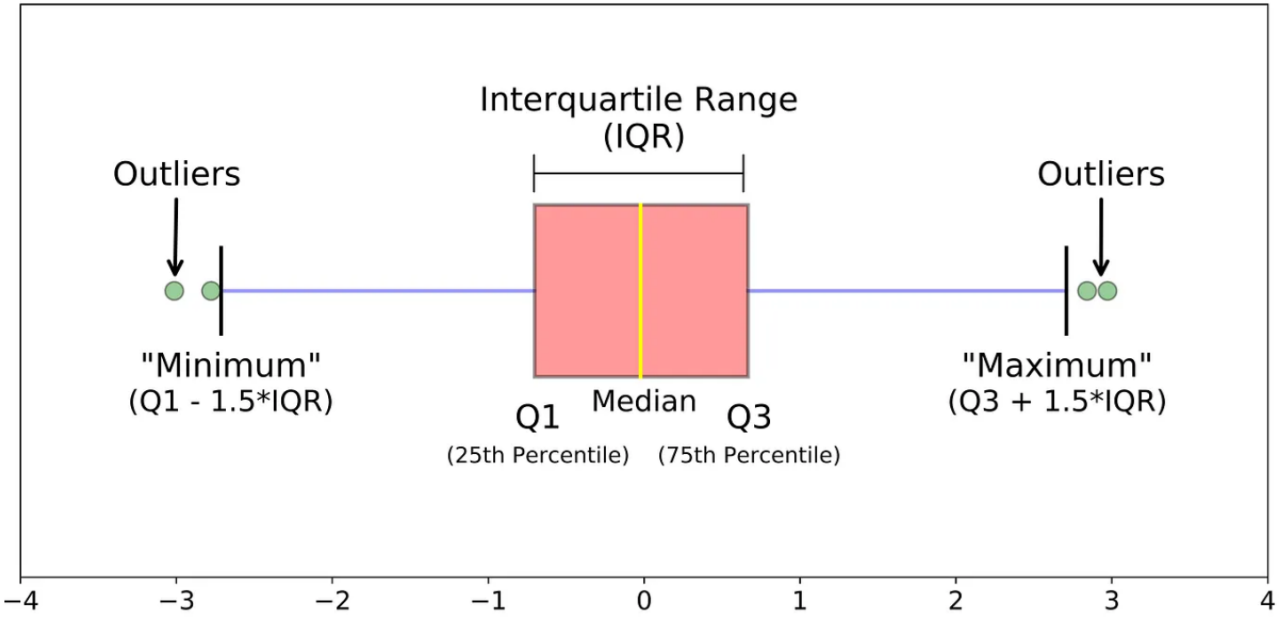
- Outliers are different from the Errors
- Outliers are interesting: It violates the mechanism that generates the normal data



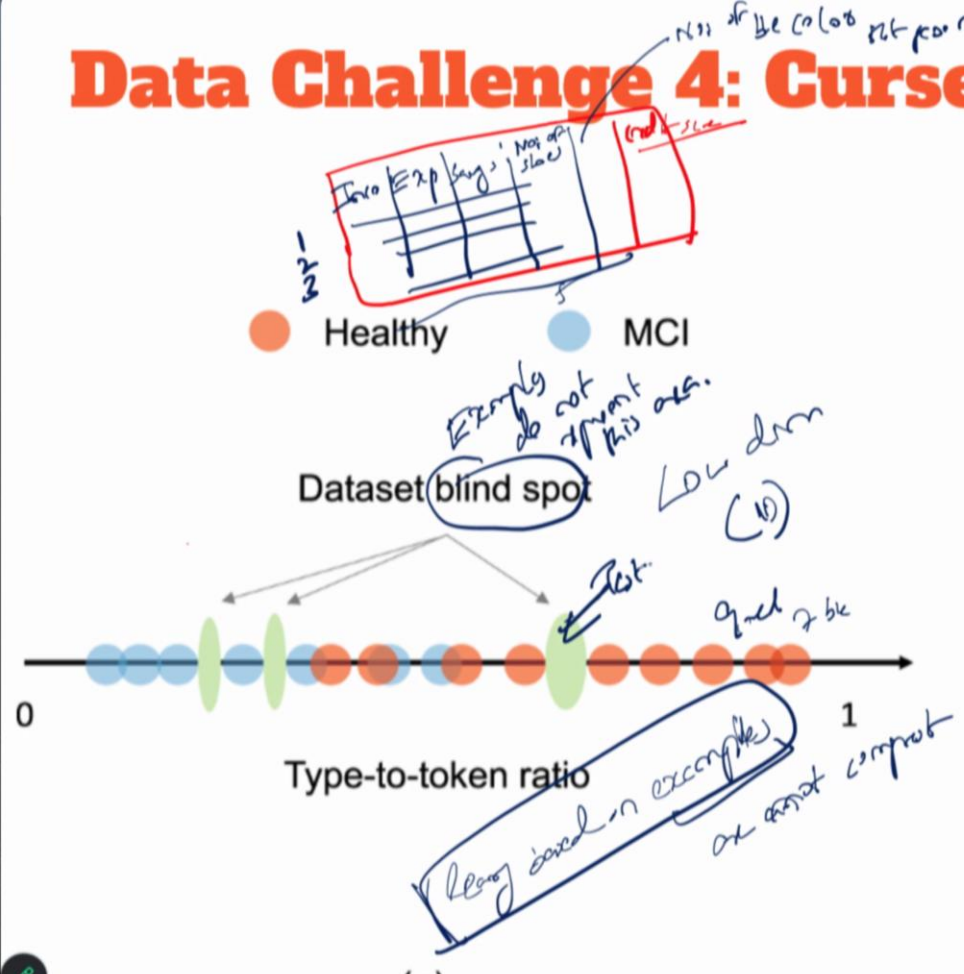
MIE = fn (of "ALL" datapoints)

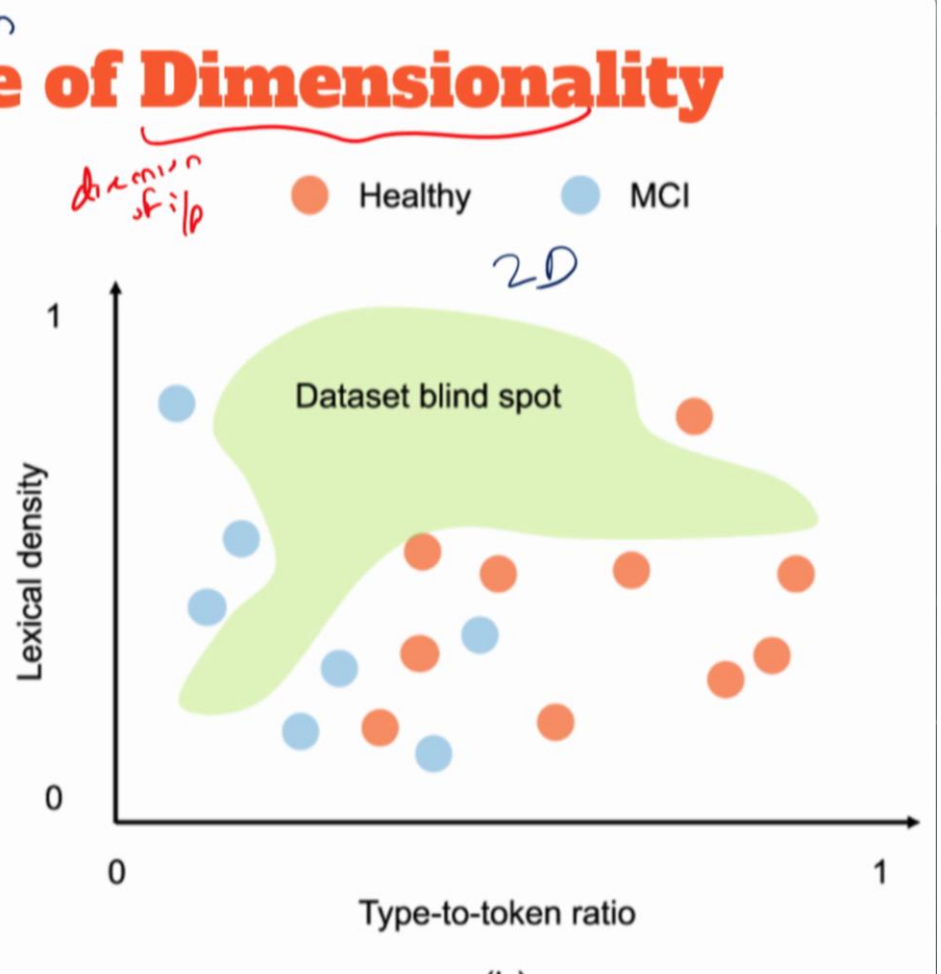
00:28:56:10
02:22:23:01


25%
25%
25%
25%
Quartile 1
Quartile 2
Quartile 3


Interquartile Range
(IQR)
Outliers
Outliers
"Minimum"
(Q1 - 1.5*IQR)
Q1
(25th Percentile)
Median
Q3
(75th Percentile)
"Maximum"
(Q3 + 1.5*IQR)
−4
−3
−2
−1
0
1
2
3
4

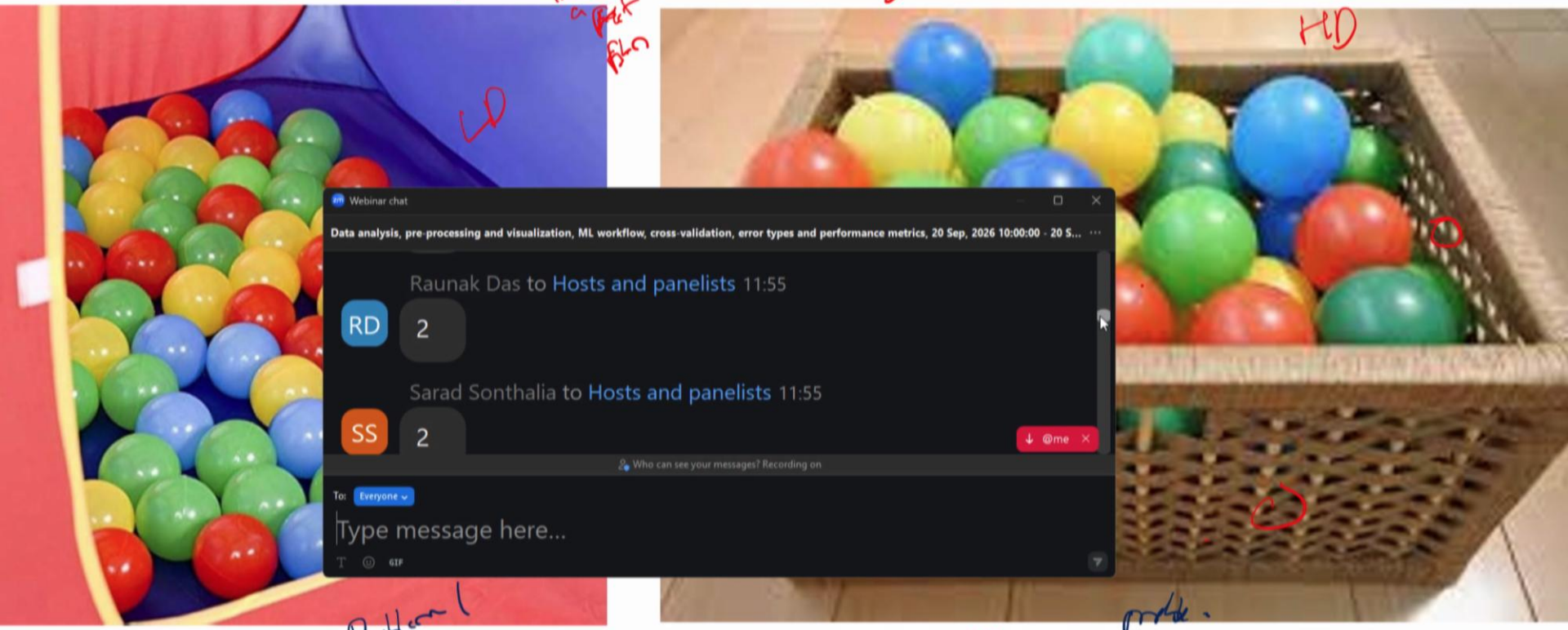
Data Challenge 4: Curse of Dimensionality





More Data Requirement

Data Challenge 4: Curse of Dimensionality

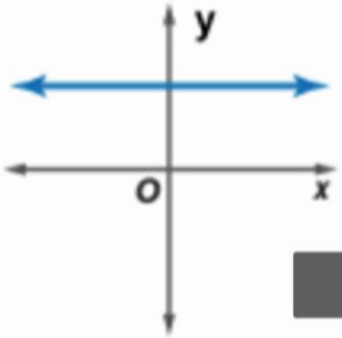


Requirement of Complex ANN Model

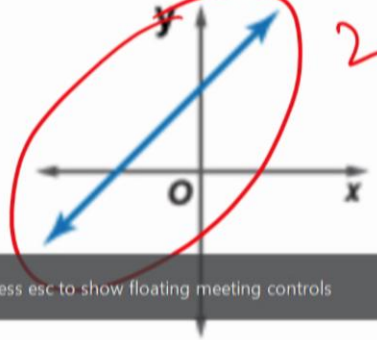
Model Challenge 1: Complexity

Adarsh Udupi +4 others raised hands View ×

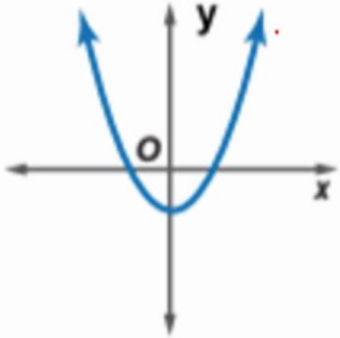
Constant function
Degree 0



Linear function
Degree 1

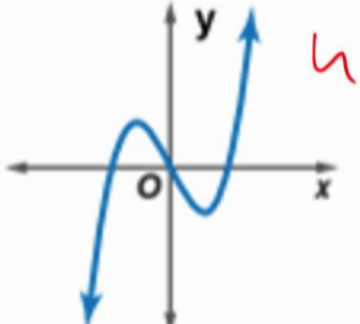


Quadratic function
Degree 2

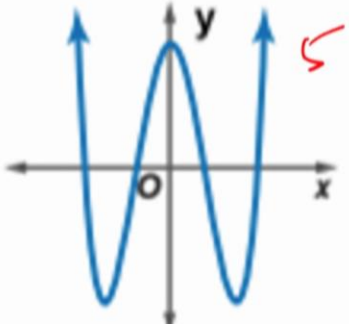


Press esc to show floating meeting controls

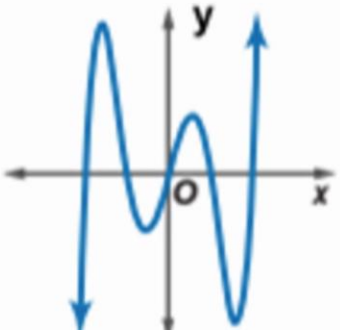
Cubic function
Degree 3



Quartic function
Degree 4

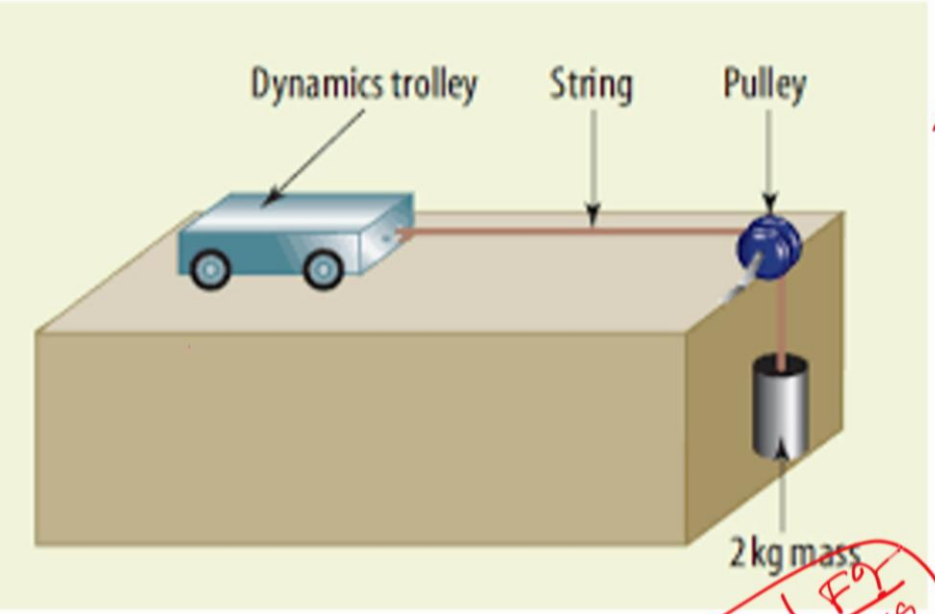


Quintic function
Degree 5



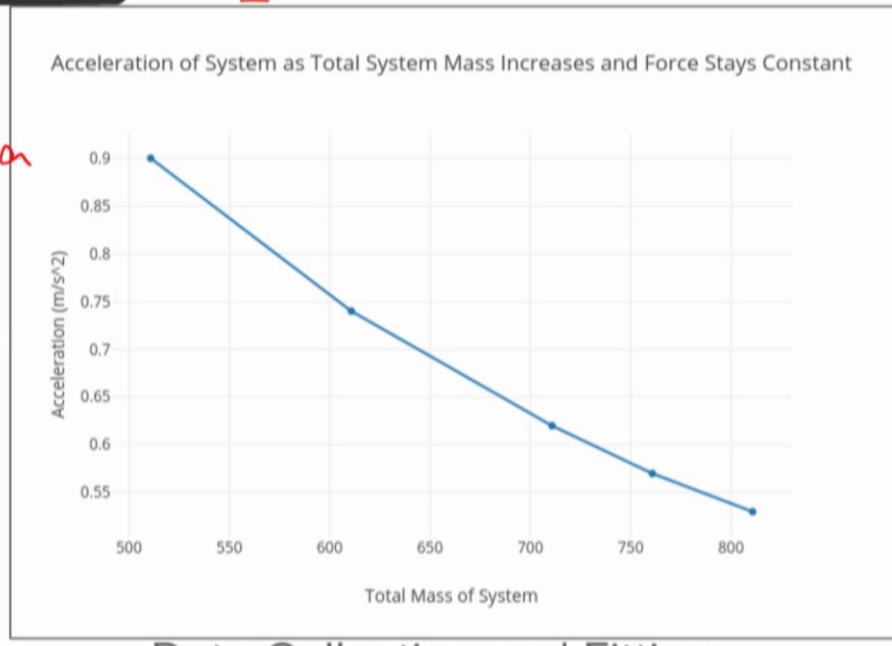
Deriving Physics Equations

Adarsh Udupi +4 others raised hands View ×



Experiment Set-up

F=ma



Data Collection and Fitting

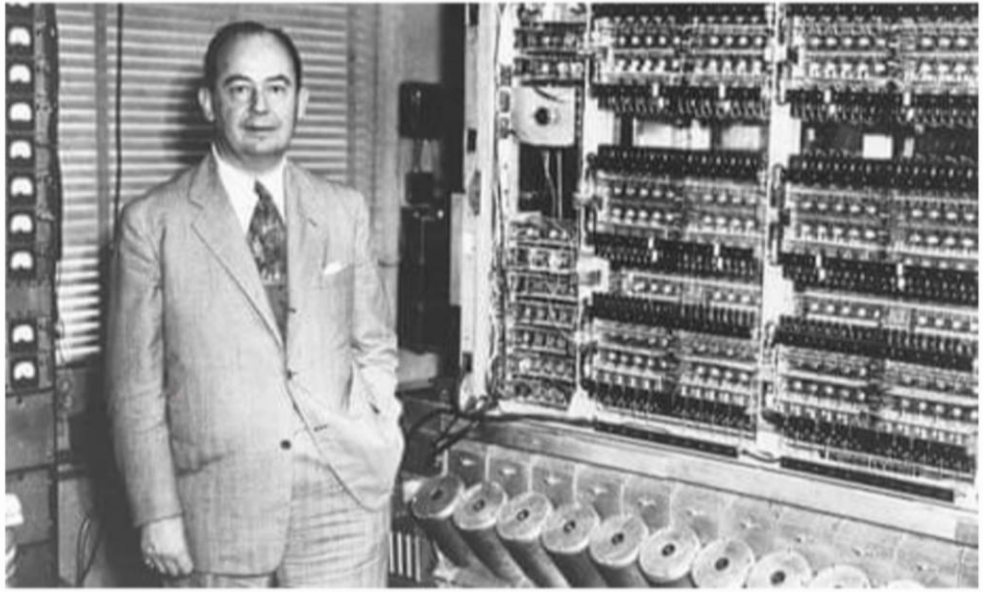
m	a	F=ma
2	3	6
1	5	5
5	2	10

Model: Acceleration = Const X 1/mass

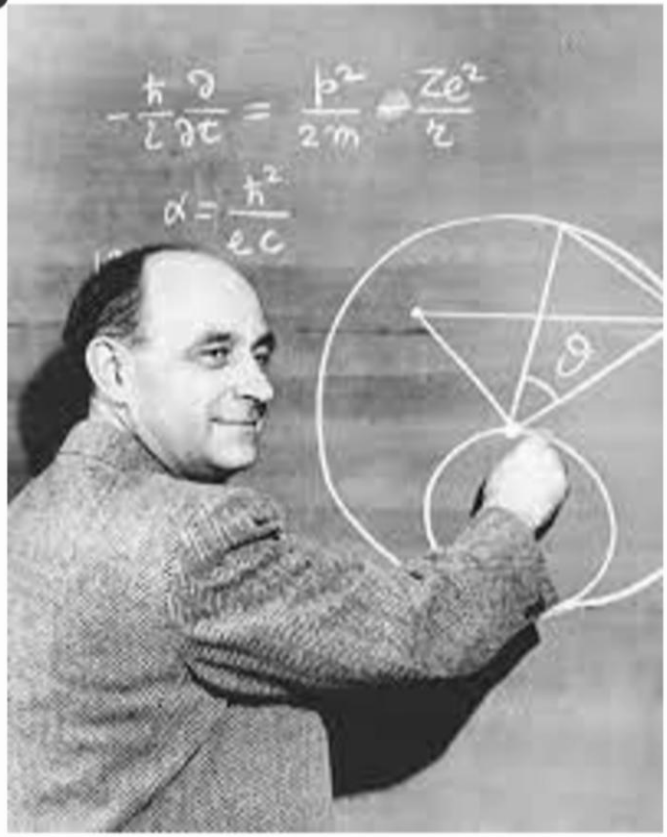
Interpretation of Quantity (Const = F)

Von Neumann about Fermi's Results

Adarsh Udupi +4 others raised hands View ×



John Von Neumann (1903-1957)
Pioneer in Digital Computer, Quantum Physics,
Game theory, etc



Enrico Fermi (1901-1954)
Pioneer in Radioactivity, Fermi-Dirac distribution, etc

Von Neumann about Fermi's Results

Adarsh Udupi +4 others raised hands View ×

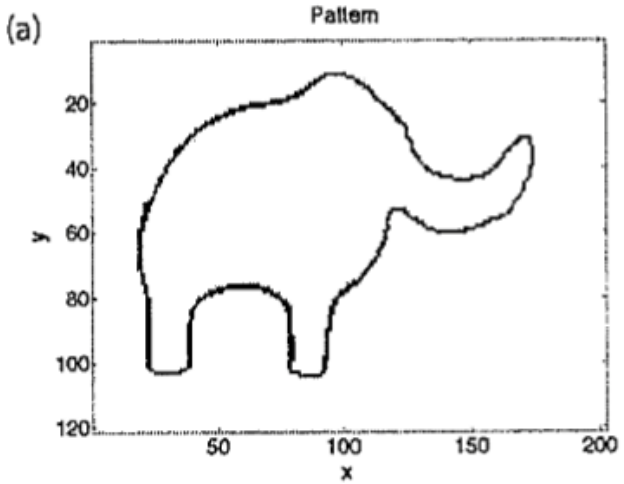
Neumann: How many arbitrary parameters have you used for calculations?
Fermi: Many (with physical interpretation)
Neumann: ***"With four parameters I can fit an elephant, and with five I can make him wiggle his trunk."***



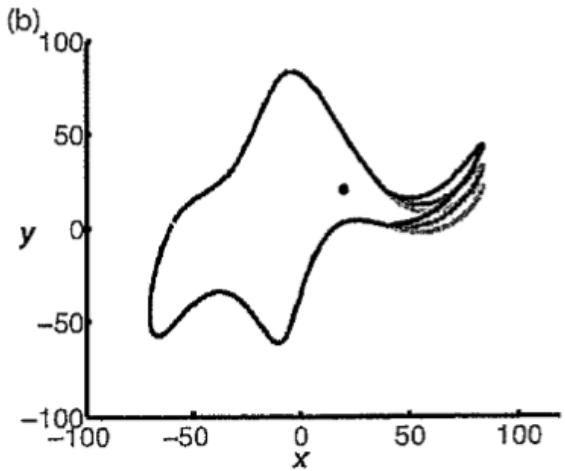
Von Neumann about Fermi's Results

Neumann: How many arbitrary parameters he used for his calculations?
Fermi: Many (with physical interpretation)
Neumann: ***"With four parameters I can fit an elephant, and with five I can make him wiggle his trunk."***

Parameter
$p_1 = 50 - 30i$
$p_2 = 18 + 8i$
$p_3 = 12 - 10i$
$p_4 = -14 - 60i$
$p_5 = 40 + 20i$



Outline of elephant (4 params)

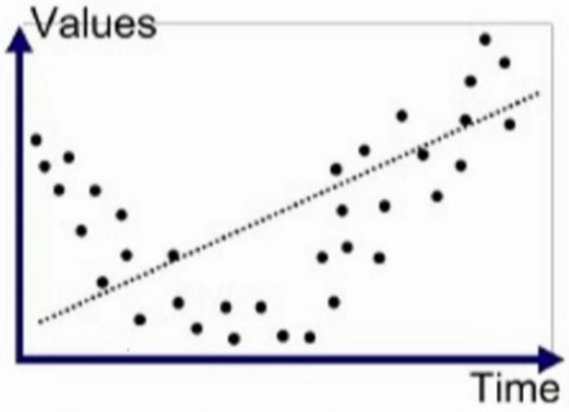


Three snapshots of the wiggling trunk (via 5 th param)

Mayer, Jurgen; Khairy, Khaled; Howard, Jonathon (May 12, 2010). "Drawing an elephant with four complex parameters". *American Journal of Phys*

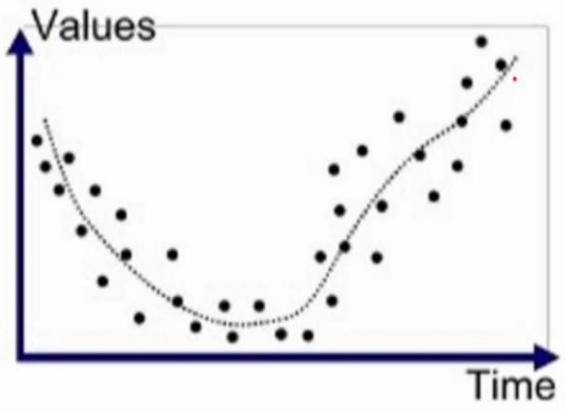
Model Challenge 1: Complexity

Adarsh Udupi +4 others raised hands View ×



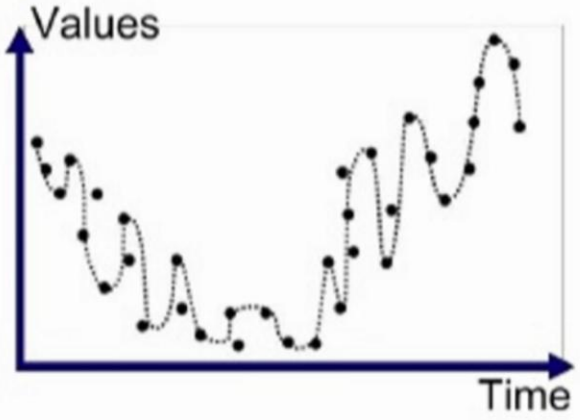
(first order polynomial)

Underfitted



(10th order polynomial)

Good Fit/Robust

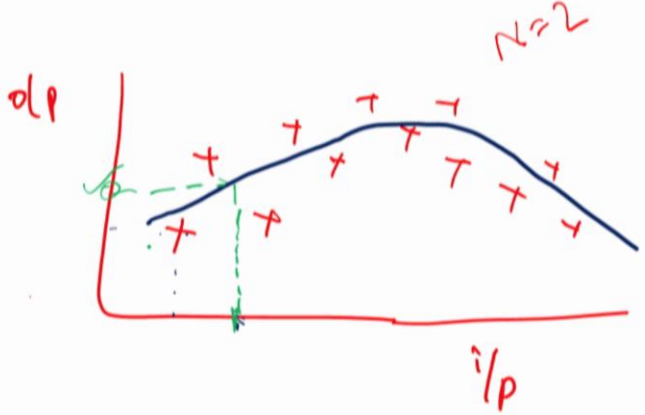


(40th order polynomial)

Overfitted

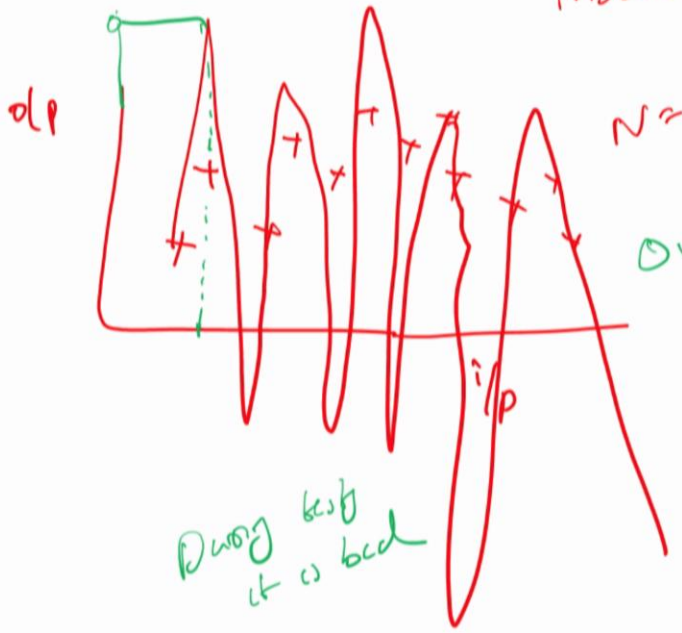


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MSE=1.2

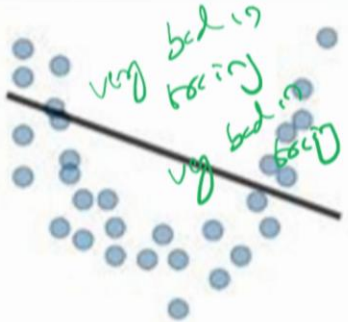
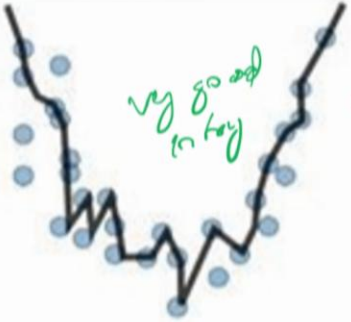
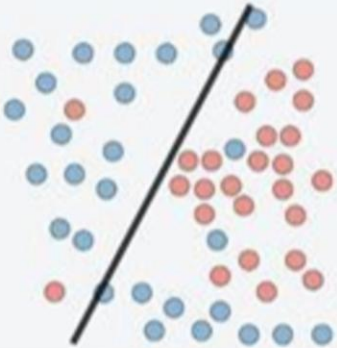
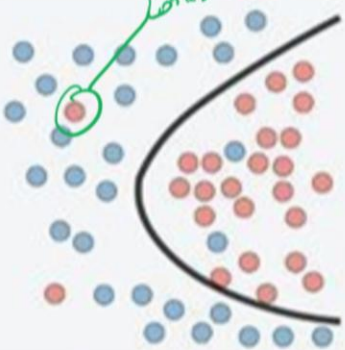
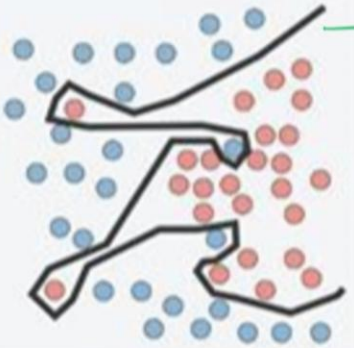
Doing testing it is better



Doing testing it is bad

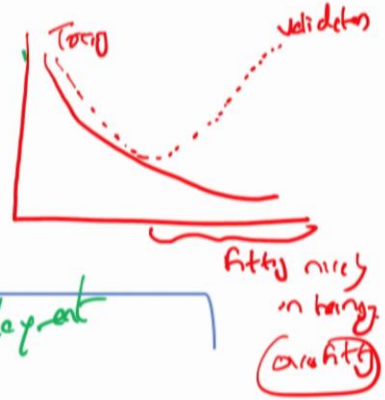
Classification vs [illegible]

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	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Regression illustration			
Classification illustration			

Training vs Validation: For Overfitting

Adarsh Udupi +4 others raised hands View ×



Available Data

Deployment

Training

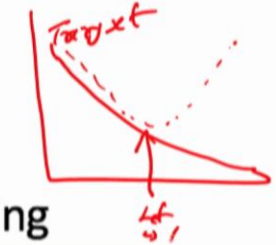
Testing

(holdout sample)

New Available Data

shrinked

Check overfitting



Training

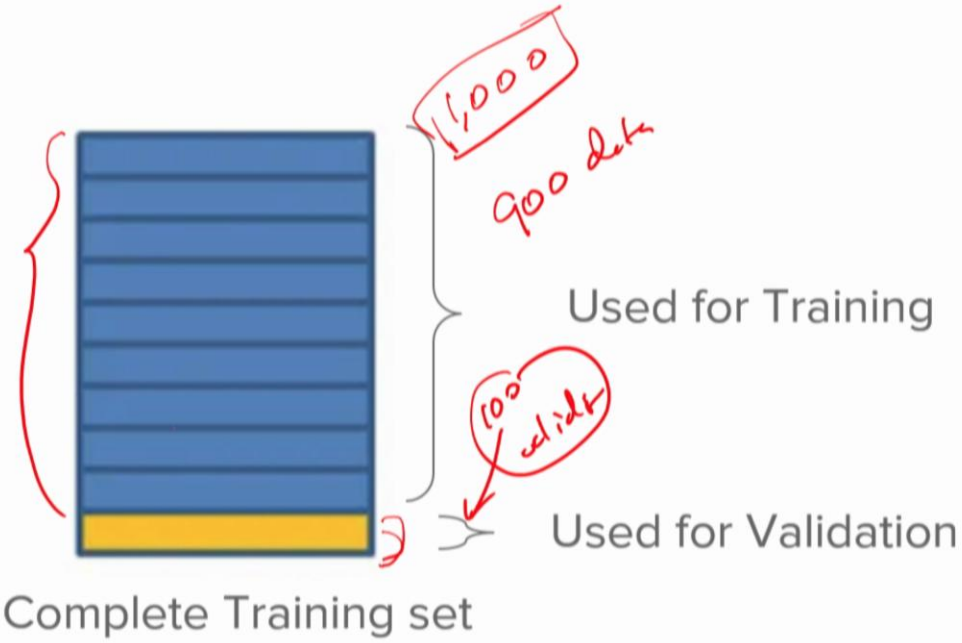
Validation

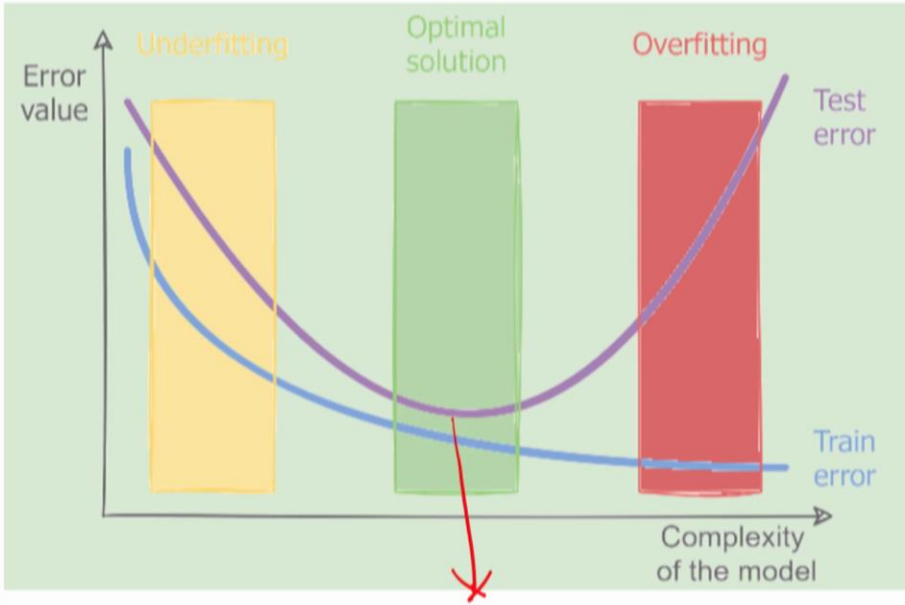
Testing

(validation holdout sample)

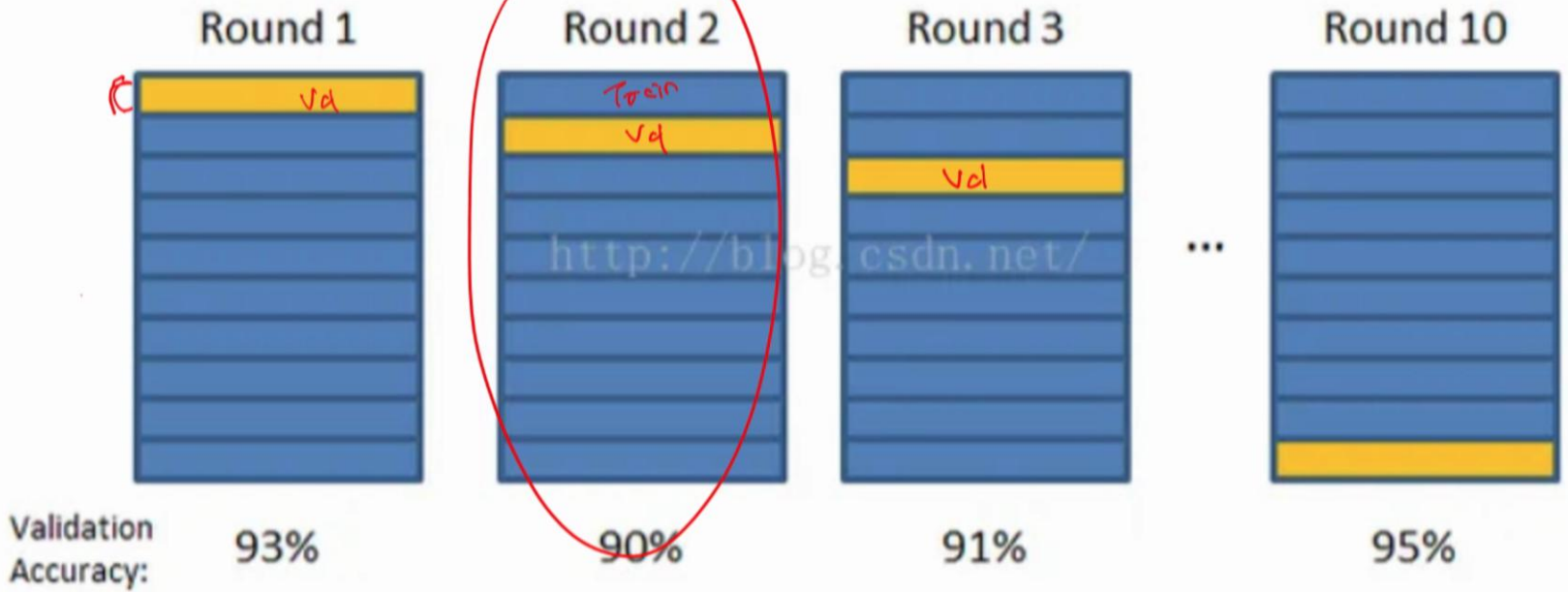
(testing holdout sample)

Training vs Validation: For Overfitting





K-fold Cross Validation



Final Accuracy = Average(Round 1, Round 2, ...)

K-fold cross validation: Train on (K–1) folds and test on 1 fold as vali

http://blog.csdn.net/

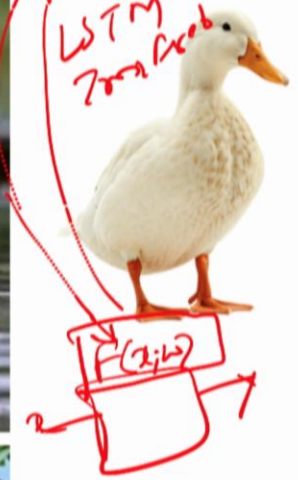
00:55:28:10 02:22:23:01

Model Challenge 2: Inductive Bias











Model Challenge 3: Evaluation



Measured or Perceived	Reality: True	Reality: False
Positive: True	Correct ☺ (TP)	Type 1 error: False Positive
Negative: False	Type 2 error: False Negative	Correct ☺ (TN)

Error Types: Case Study



Cancer Diagnostics



Anti-missile System



Weather Predictions

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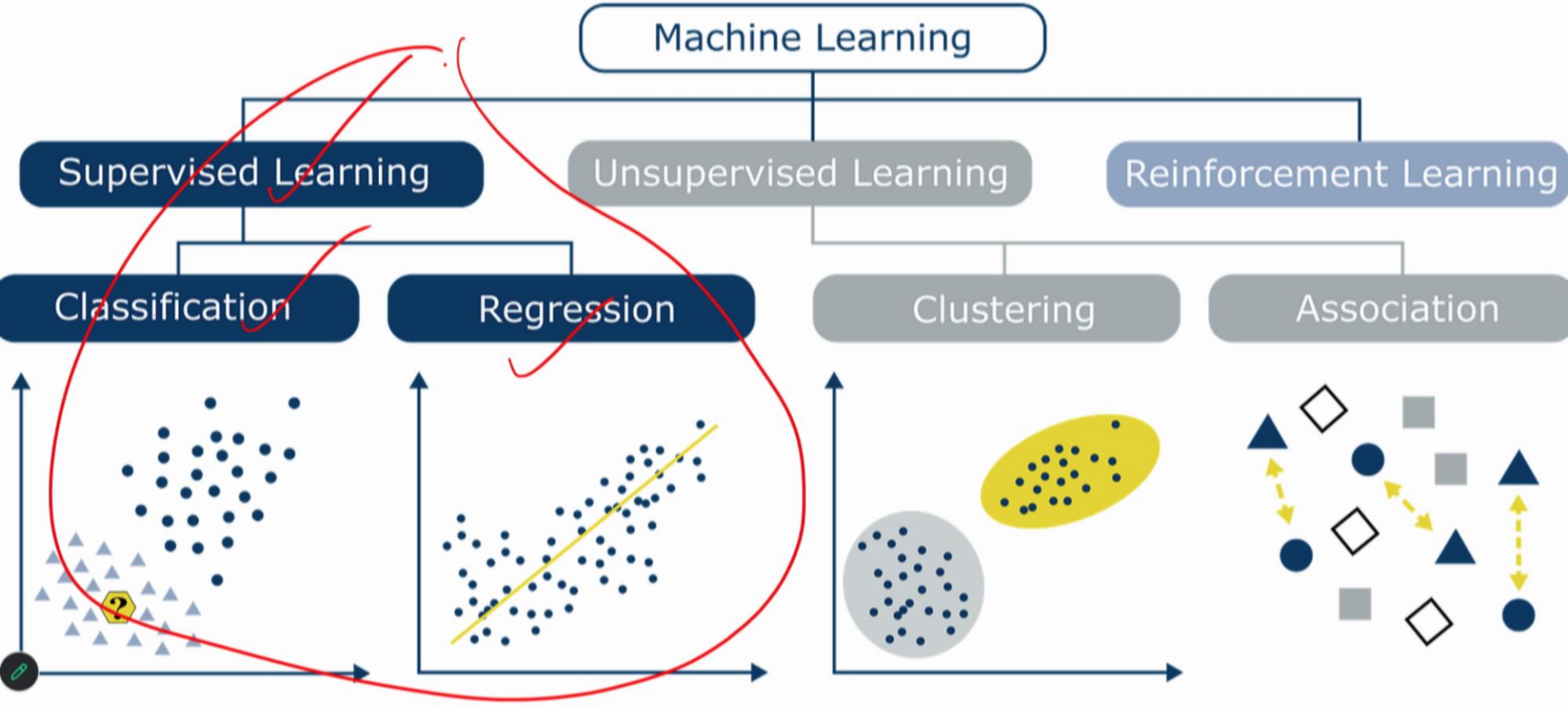
Performance Metrics

Key: **tp** = True Positive, **tn** = True Negative, **fp** = False Positive, **fn** = False Negative

Metric Name	Metric Forumla	When to use
Accuracy	**Accuracy** $= \frac{tp + tn}{tp + tn + fp + fn}$	Default metric for classification problems. Not the best for imbalanced classes.
Precision	**Precision** $= \frac{tp}{tp + fp}$	Higher precision leads to less false positives.
Recall	**Recall** $= \frac{tp}{tp + fn}$	Higher recall leads to less false negatives.
F1-score	**F1-score** $= 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$	Combination of precision and recall, usually a good overall metric for a classification model.
Confusion matrix	NA	When comparing predictions to truth labels to see where model gets confused. Can be hard to use with large numbers of classes.

Outline

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Introduction to Unsupervised Learning











VS



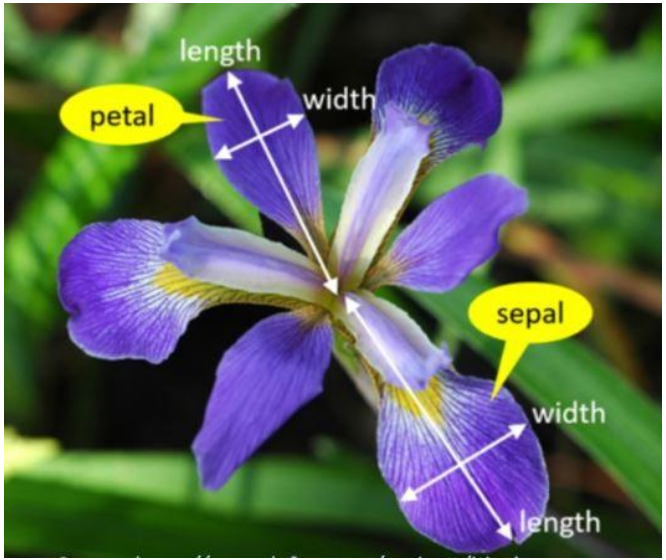

Set of good actions
Policy
My copy place
262

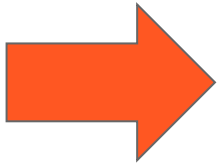
Clustering

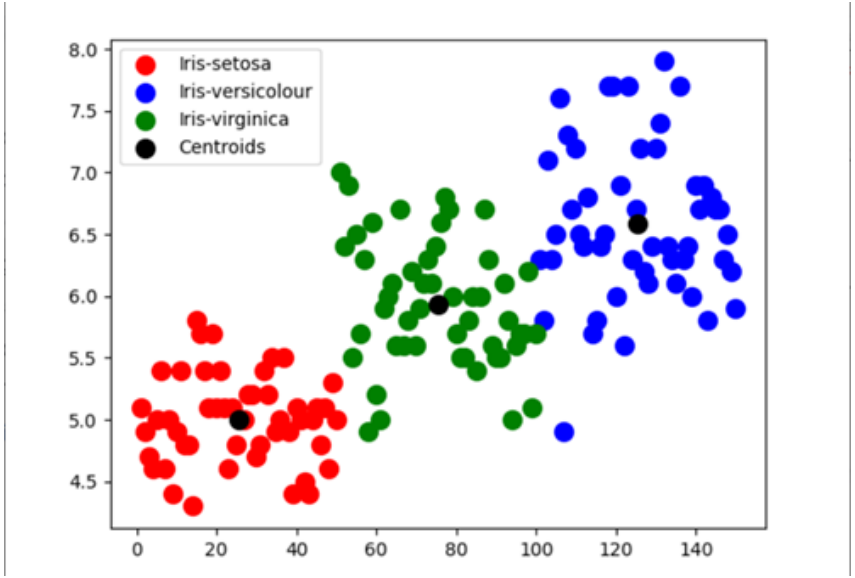

Versicolor


Setosa

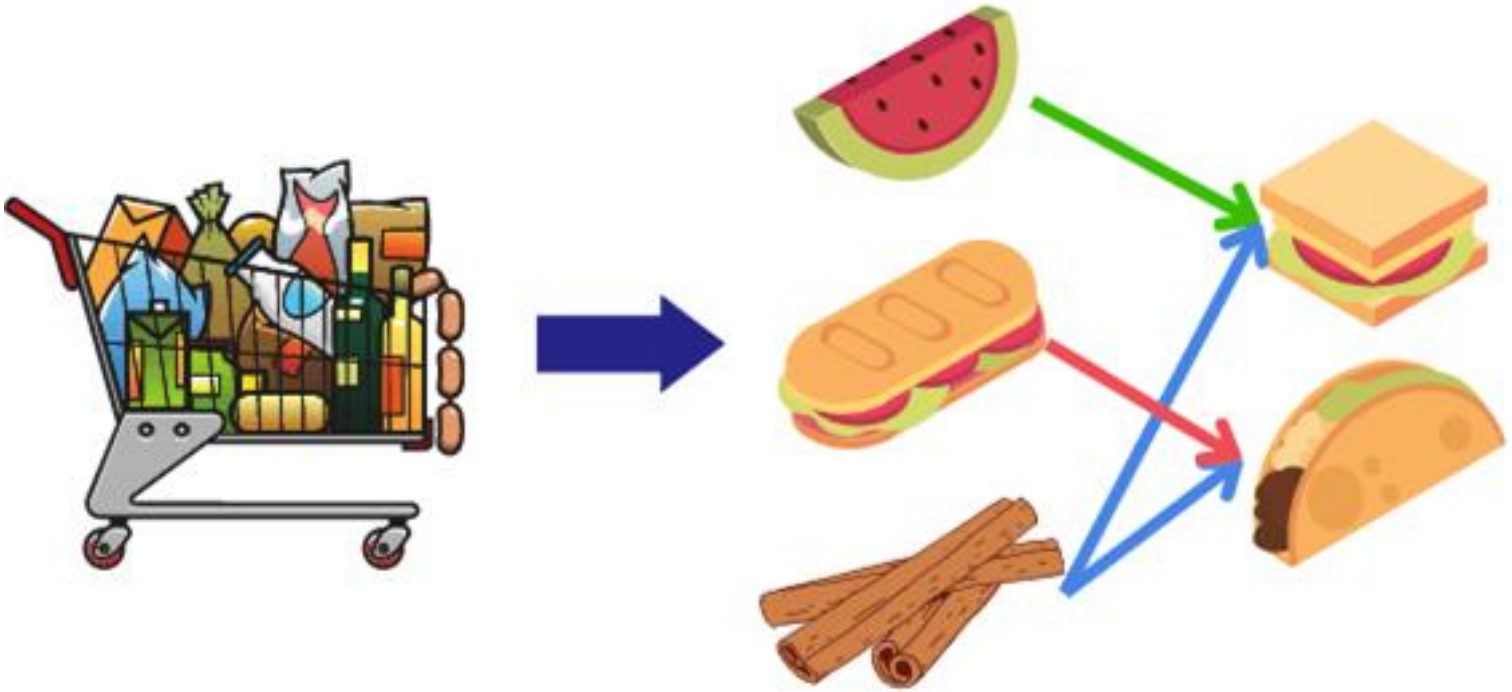

Virginica


length
petal
width
sepal
width
length




Iris-setosa
Iris-versicolour
Iris-virginica
Centroids
8.0
7.5
7.0
6.5
6.0
5.5
5.0
4.5
0
20
40
60
80
100
120
140

Association

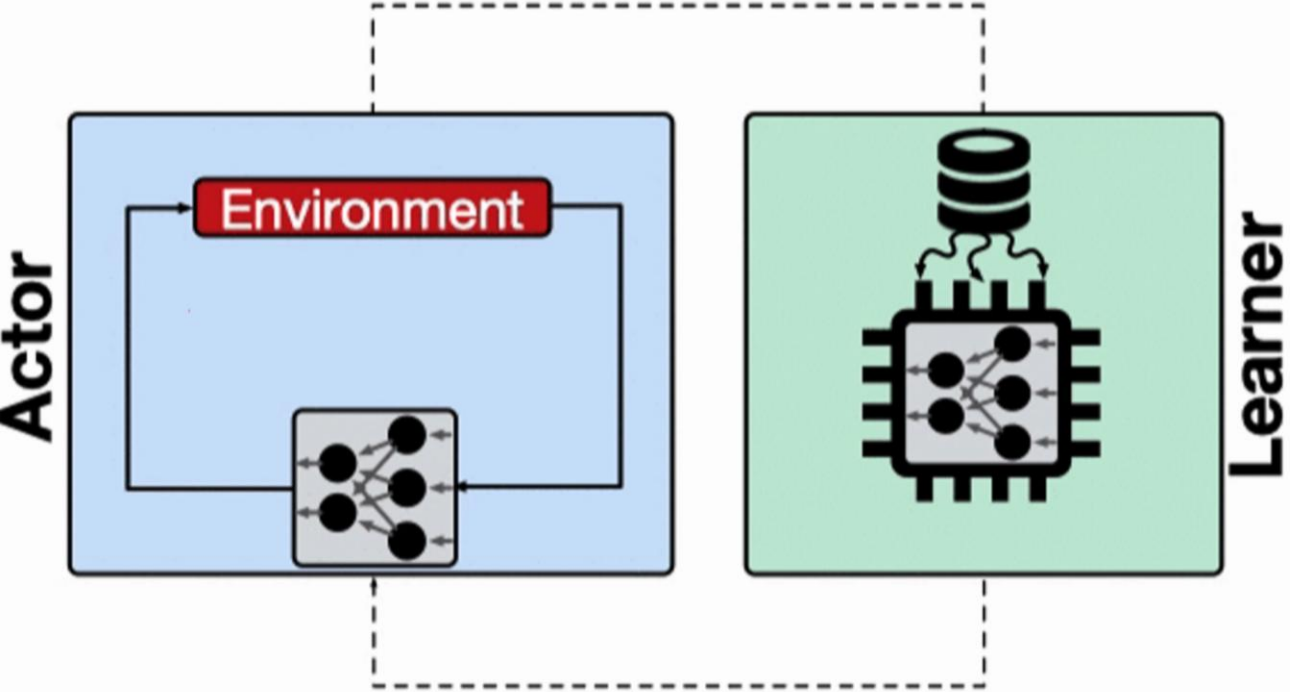


"93% of people who purchased item A also purchased item B"

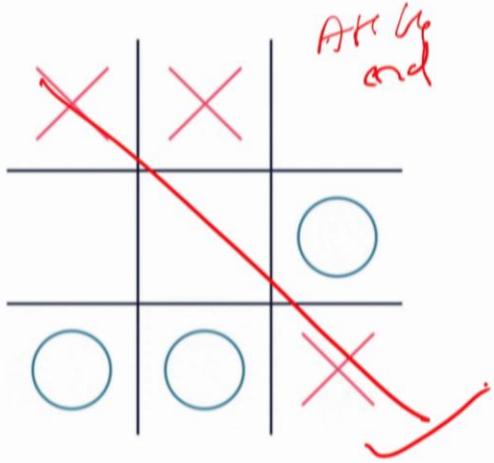
Introduction to Reinforcement Learning


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Reinforcement Learning



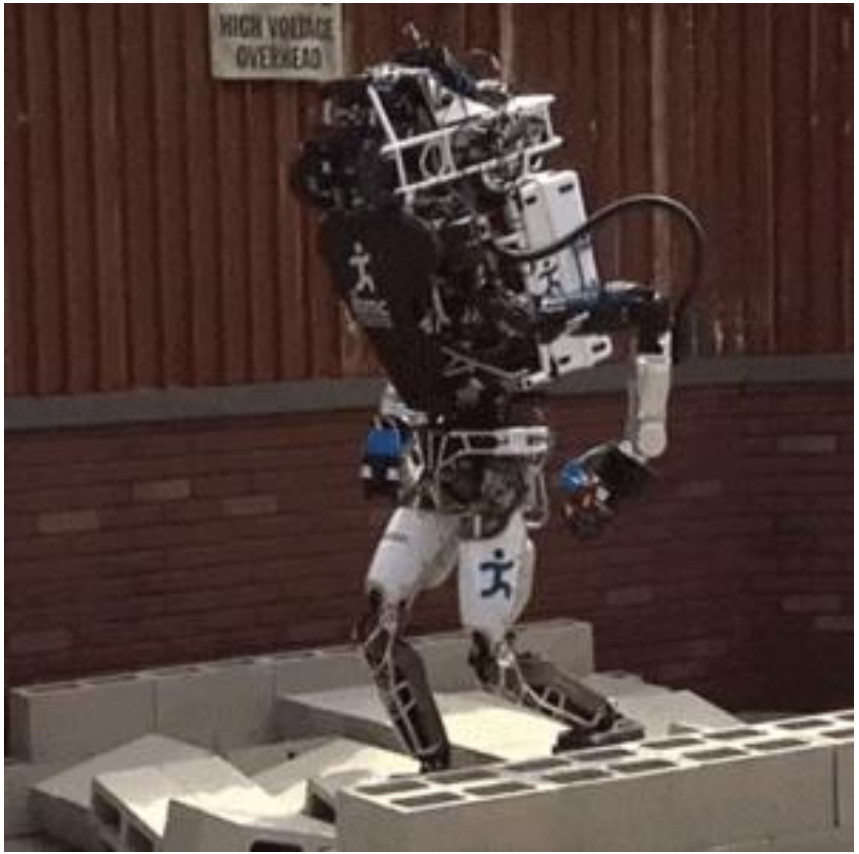






Reinforcement Learning

Objective: Move forward while minimizing how often it fails and the energy it expends.



Learning Phase



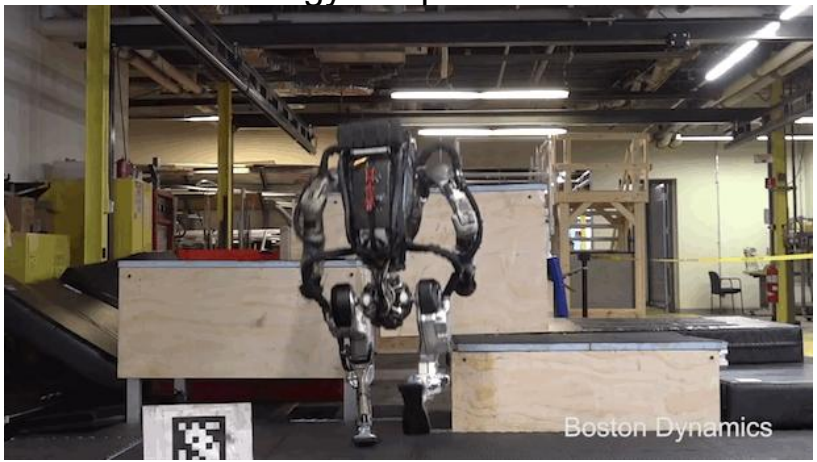


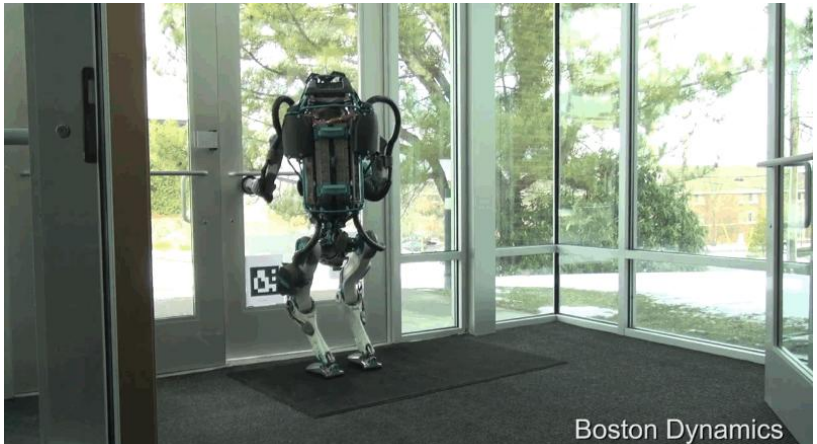
Reinforcement Learning

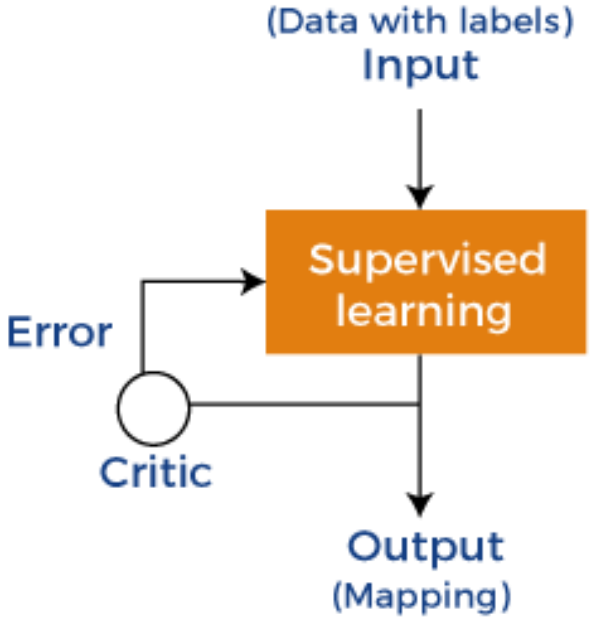
Objective: Move forward while minimizing how often it fails and the energy it expends.

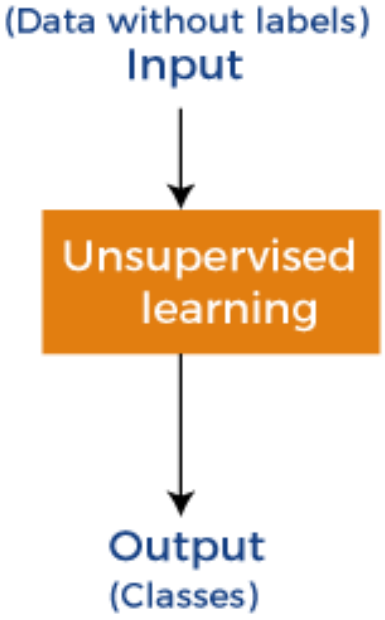


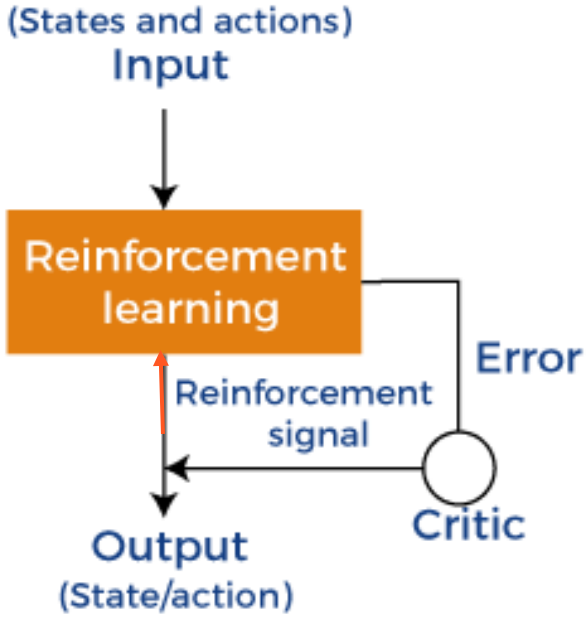
Learning Phase






(Data with labels)
Input
Supervised learning
Error
Critic
Output
(Mapping)


(Data without labels)
Input
Unsupervised learning
Output
(Classes)


(States and actions)
Input
Reinforcement learning
Error
Reinforcement signal
Critic
Output
(State/action)







Industry-relevant Exercise

